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# Multi-vehicle formation control and obstacle avoidance using negative-imaginary systems theory

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ABSTRACT

This paper presents a compelling combination of the two negative-imaginary (NI) control systems: a consensus-based formation control framework for a hybrid multi-vehicle system and a new dynamic obstacle detection and avoidance algorithm. The incorporation of the two techniques permits robots to self-detect collisions and then self-create a safe path to overcome any unexpected obstacles while traveling to the destination in a given formation. Moreover, owing to the distributed and dynamic architecture, our NI obstacle avoidance method has better adaptability to the change of environment conditions or obstacle/robot dynamics. Also, a rigorous comparative study on the performance of our obstacle avoidance method with respect to that of the artificial potential field function and switching formation-based NI obstacle avoidance algorithms is performed as a benchmark. Finally, simulation and experimental results are then implemented to demonstrate the effectiveness of the proposed theories.
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## 1. Introduction

Multi-vehicle systems have drawn growing attention from researchers owing to the diversity of the applications they can perform. Examples include monitoring, surveillance, product delivery, self-localization and mapping (SLAM), and disaster response. To implement these real-time applications, autonomous vehicles must work together via an information exchange protocol to achieve desired goals. For this collaboration to be effective, various formation control architectures and strategies (Dong, Zhou, Ren, & Zhong, 2017; Ortega, Muñoz, Quesada, Garcia, & Ordaz, 2015) have been explored by numerous researchers using three primary formation methods emerging: virtual structure and behavior (Guerrero, Fantoni, Salazar, & Lozano, 2010), virtual leader, and leader–follower (L–F) (Hou & Fantoni, 2015; Mercado, Castro, & Lozano, 2013; Tran, Garratt, & Petersen, 2020a). Among these basic strategies, the L–F cooperative structure has been most frequently used due to its simplicity and effectiveness in control and implementation.

Although many various cooperative methods based on the L–F formation strategy have been widely developed in the literature (Guerrero, 2011; Guney & Unel, 2013; Hou & Fantoni, 2015) and now there are many ways to deal with different aspects of the nominal consensus and cooperative control problem, robust cooperative control is still less studied. In Dong, Li, Zhao, and Ren (2016), Dong et al. (2017) and Seo, Kim, Kim, and Tsourdos (2012), stable formation tracking control has been verified, but the internal stability of the networked robots has not been considered. In addition, existing results on robust cooperative control of heterogeneous multi-agent systems are restricted to either second order plants/controllers, or minimum phase linear time-invariant (LTI) systems or virtual string connections (Dong et al., 2017; Su & Huang, 2014; Wang, Dong, Li, & Ren, 2018; Zhu & Chen, 2014) while complicated robot systems such as unmanned aerial and ground vehicles (UAVs and UGVs) should be modeled at a higher order for accurate representation according to Li and Li (2011) and Zhang, Song, Qiao, Zhang, and Peng (2014).

In order to overcome these shortages and diversify the L–F cooperative control applications, negative-imaginary (NI) systems consensus theory has been presented. This theory is a class of robust control that yields the robustness of the whole consensus-based cooperative control architecture associated with multi-robot systems containing various high order robot dynamics in the presence of external disturbances and additive NI model uncertainty (Petersen & Lanzon, 2010; Song, Lanzon, Patra, & Petersen, 2012; Wang, Lanzon, & Petersen, 2015). The theory (Wang et al., 2015) states that a robust output feedback consensus for multi-agent systems with undirected and connected network communication graphs can be obtained if the three following

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elements exist: (1) a consensus output feedback; (2) heterogeneous strictly-negative-imaginary (SNI) single input–single output (SISO) plants or heterogeneous multi input–multi output (MIMO) plants; and (3) NI controllers.

Simulation results in Wang et al. (2015) including external disturbances and model uncertainty have shown that cooperative output tracking of networked SNI/NI systems can be performed efficiently while all agents are driven from various initial positions to the final rendezvous point. However, Wang et al. (2015) only introduced an NI-systems consensus control framework whose practical applications are still fairly limited, for example, NI systems are unable to follow a predefined trajectory in a given spatial pattern. Therefore, in our previous work (Tran, Garratt, & Petersen, 2017), we designed a leader–follower NI consensus-based formation control protocol based on the NI consensus theory. The last experimental results indicated that a pair of UAV systems was able to keep horizontal-line formation and track their trajectory. Compared to the previously published approaches on formation control for swarm systems using the global information (Dong et al., 2017; Wang et al., 2018; Zhang et al., 2014), the significant contributions of our NI formation control approach are fourfold: (1) formation patterns can be maintained even at a small size by maintaining the relative distances between agents; (2) only small amounts of data transmission and computation resources are processed since the expected formation is generated via relative distance information measured directly by machine vision systems; (3) the position consensus problem that drives all agents to agree on a majority value can be studied as an internal stability one between the NI systems and their controllers to achieve the desired position; and (4) the NI formation control protocol and the additional NI control algorithms may be well combined together while the general stability properties of all control loops are still guaranteed.

Now, the NI invariant consensus-based formation algorithm alone is not sufficient to implement the desired tasks in a cluttered environment where obstacle avoidance behavior is extremely critical. Whilst existing methodologies such as artificial potential field (APF) (Ge & Cui, 2002; Khatib, 1986), enhanced APF, null-space-based behavioral control algorithm (Antonelli, Arrichiello, & Chiaverini, 2008), or D* Lite algorithm (Ganapathy, Yun, & Chien, 2011) have been successfully implemented for a single UAV/UGV, they have not been extended to multi-vehicle applications due to their complicated algorithmic structure and the lack of an information communication strategy between the robots. Moreover, control stability of the expanded architectures has been undetermined. These deficiencies have been partly overcome by recent studies which address the cooperative problem and obstacle avoidance planning simultaneously for the multi-robots system. Many methods developed by researchers utilize a geometry based formation varying approach to modify the formation shape (Chen & Sun, 2016; Dai, Lee, Kim, & Wee, 2014; Lee & Chwa, 2018). Nevertheless, there are many issues need to be considered. Oscillations may exist in task-switches due to control constraints (Dai et al., 2014; Liao et al., 2017). Also, some different control laws only operate in specific scenarios or partially known environment such as avoiding either only one obstacle or two facing obstacles (Sgorbissa & Zaccaria, 2012). Furthermore, in existing work, significant controller parameters within the consensus algorithms are not adaptive, for instance, the fixed width of each routing layer in Lee and Chwa (2018). Finally, an important issue in the formation control with collision avoidance is that a suitable strategy is required to let all robots move into a given formation, while still guaranteeing collision avoidance. This extra method naturally rises general questions relating to the complexity and synchronization problems in control structure.

Addressing the aforementioned problems, we propose a novel solution using NI based obstacle detection and collision avoidance, which can effectively connect with the available NI formation approach. Considering obstacle avoidance, the proposed technique works for both stationary and moving obstacles by varying the robot's repulsive force whose magnitude is based on the robot direction and the obstacle center. Considering obstacle detection, each robot can self-check for collision via our line-circle intersection approach where the line is termed the $\overrightarrow{ahead}$ vector of the robot, and a virtual circle which envelops the geometric extent of the obstacle. When a risk of crash is detected, an avoidance force is generated to push the robot away from the obstacle. The magnitude of this force is calculated based on the amount of overlap between the robot heading vector and the obstacle's virtual circle.

Unlike other existing approaches, which are difficult to integrate with the available NI consensus-based formation control structure, our obstacle avoidance controller is able to be easily incorporated, and at the same time the robustness properties of the complete system, consisting of NI obstacle avoidance, NI formation, and SNI robot dynamics, are still ensured via a homogeneous NI framework. Compared to some recent strategies which consider just a specific scheme or partially known environments (Dai et al., 2014; Panagou & Kumar, 2014), our method is novel and effective.

In outdoor cooperative scenarios, where the UAV and UGVs are tasked to construct a complete map of unknown environments, situations may occur when an absolute position sensor (like GPS) is only available to one vehicle at a time. Because the absolute position navigation may not be accessed by all vehicles (e.g. GPS interference caused by urban canyon effect), we assume that only the relative distances between vehicles and between vehicles and obstacles can be measured with a sensor such as laser rangefinder or machine vision system. In our indoor experiments, a Vicon Motion Capture System (VMCS) imitates this information. The primary idea is that we collect the global robot positions by VMCS and subtract the known locations of obstacles and other robots to calculate a relative distance that is provided to each robot. Moreover, a decentralized control scheme without central controller is also designed to overcome increased complexity in collaborative large-scale multi-agent systems. To improve the performance of distributed robots, both proposed methods are distributed and implemented in the single board computer systems on each robot. Interaction between robots includes only binary commands and the desired location for the master that are sent at a specific time as shown in Section 5.

In summary, the contributions of this study are listed as follows:

- We are the first to provide an NI homogeneous control approach for both existing problems by generating the expected formation and avoiding all two kinds of obstacles, which is one of the main obstacles of several multi-robot control methods (Dai, Kim, Wee, Lee, & Lee, 2015; Dai et al., 2014).
- We develop a comprehensive and effective obstacle avoidance approach for the two cooperative systems (UAV–UGVs and UGVs) moving in formation.
- We create robust stability for the distributed formation control, static or mobile obstacle avoidance problem, and dynamics of agents via the NI stability criterion.
- We conduct extensive comparative studies in real tests as well as in numerical simulations to highlight the efficacy of the proposed obstacle avoidance system in terms of avoiding local minima traps and producing smooth obstacle avoiding

trajectories, compared to the performance of the APF control technique and the switching formation-based NI obstacle avoidance (NI-SFOA) strategy (Tran, Garratt, & Petersen, 2020b).

The rest of this paper is organized as follows: Preliminary descriptions of graph theory and NI formation architecture are reviewed in Section 2. UAV and UGV dynamics are mentioned in Section 3. Section 4 presents the obstacle detection and avoidance using NI theory, linked with the NI formation control structure and a cooperative strategy in Section 5 and Section 6 respectively. The stability of complete system is verified in Section 7, while Section 8 introduces the hardware/software configuration and architecture design for the experiments on formation and collision avoidance capability, Section IX provides numerical simulations and the real experiments, followed by a brief conclusion drawn in Section 10.

## 2. Preliminaries of graph topology and negative-imaginary theory for the formation control

### 2.1. The basics of graph theory

Graph theory can be used to effectively describe the inter-agent communication including the number of unmanned robots $N$. Each robot is represented as a node. Let $G \in \mathbb{R}(N \times L) = (\vartheta, \varepsilon)$ denote a directed graph with $N$ nodes and $L$ edges, such that $\vartheta = (v_1, v_2, \ldots, v_N)$ and $\varepsilon \subseteq \{(v_i,v_j)\colon v_i,v_j \in \vartheta,\ v_i \neq v_j\}$ mathematically describe the finite nodes set and the ordered edges set, respectively. Suppose that each edge of $G$ is assigned an orientation, which is arbitrary but fixed.

#### 2.1.1. The incidence matrix

The incidence matrix of $G$, denoted by $\mathcal{Q}_a$, is a $N\times L$ matrix defined as follows:

$$\mathcal{Q}_a = \begin{cases} \mathcal{Q}_{a_{ij}} = 1 & \text{if } v_i \text{ is the head of edge } e_{v_i,v_j}, \\ \mathcal{Q}_{a_{ij}} = -1 & \text{if } v_i \text{ is the tail of edge } e_{v_i,v_j}, \\ \mathcal{Q}_{a_{ij}} = 0 & \text{if } v_i \text{ is not connected to } e_{v_i,v_j}. \end{cases} \tag{1}$$

Note that the incidence matrix has a column sum which equals to zero, which is assumed from the fact that every edge must have exactly one head and one tail. Extending this graph theory, given an arbitrary oriented set of the edge $\varepsilon$, the graph Laplacian, $L(G)$, is defined as $L(G) = \mathcal{Q}_a(G)\mathcal{Q}_a(G)^T$.

For convenience in interpreting further the UAV–UGV/UGVs cooperative theory, some mathematical information graph matrices are newly described.

#### 2.1.2. The consensus matrix

Let the $N\times L$ matrix $\mathcal{Q}_b$ represent a consensus matrix with $N$ nodes and $L$ edges which corresponds to a graph $(\vartheta_b, \varepsilon_b)$. This $N\times L$ matrix indicates which robot at node $v_i$ in $\vartheta_b$ will send its consensus information to the adjacent nodes $v_j$ in $\varepsilon_b$.

$$\mathcal{Q}_b = \begin{cases} \mathcal{Q}_{b_{ij}} = 1 & \text{if } v_i \text{ is the head of } e_{v_i,v_j} \text{ for consensus}, \\ \mathcal{Q}_{b_{ij}} = -1 & \text{if } v_i \text{ is the tail of } e_{v_i,v_j} \text{ for consensus}, \\ \mathcal{Q}_{b_{ij}} = 0 & \text{otherwise}. \end{cases} \tag{2}$$

#### 2.1.3. The reference matrix

The reference matrix $\mathcal{Q}_r \in \mathbb{R}(N \times L)$ indicates the node or the UAV/UGV in a collaborative group that will follow specific trajectories:

$$\mathcal{Q}_r = \begin{cases} \mathcal{Q}_{r_{ij}} = 1 & \text{if } v_i \text{ directly follows the reference path}, \\ \mathcal{Q}_{r_{ij}} = 0 & \text{if } v_i \text{ does not directly follow the reference path}. \end{cases} \tag{3}$$

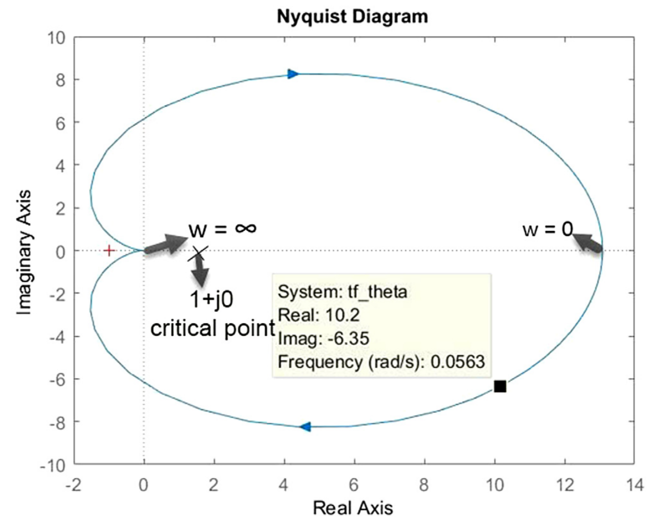



**Fig. 1.** Nyquist plot of the SNI loop transfer function L(s) = M(s)N(s).

### 2.2. Negative-imaginary theory for the formation control

According to Petersen and Lanzon (2010), the definition of an SNI or NI system is recalled in this section. A SISO transfer function $P(s)$ is SNI if its Nyquist plot is located absolutely below the real axis for all positive frequencies and all poles of $P(s)$ are located in the left-half plane. The figure of this plot starts from an arbitrary point on the real axis at the frequency $\omega = 0$ and ends at the center point $(0, 0)$ at $\omega = \infty$. This property is mathematically derived as follows: $P(s) \in SNI$ if $j[P(j\omega)-P^*(j\omega)] > 0$ for all $\omega > 0$. Being less strict than the SNI system, the shape of the NI Nyquist plot is similar to that of the SNI system; however, its figure is allowed to touch, but not allowed to cut the real axis; i.e., $P(s) \in NI$ if $j[P(j\omega)-P^*(j\omega)] >= 0$.

**Lemma 2.1.** *A positive connection between an NI system and an SNI system results in an SNI systems structure (Petersen & Lanzon, 2010).*

In order to control and stabilize this particular system, the theory (Petersen & Lanzon, 2010) indicates that a positive feedback interconnection between an SNI or NI plant $M(s)$ and its NI or SNI controller $N(s)$ is internally stable if and only if the loop gain transfer function at zero frequency is strictly less than 1; notably, $M(0)N(0) < 1$ (Fig. 1).

For MIMO NI systems, a multi-agent consensus approach was developed in Wang et al. (2015), followed by an enhanced NI-systems formation control architecture based on this consensus method (Tran et al., 2017). In the former technique, given a square SNI transfer function matrix $P(s)$, a communication graph $G$, an incidence matrix $\mathcal{Q}$, and an NI controller matrix $\bar{P}(s)$, a distributed output feedback control law obtains robust output feedback consensus as given by the formulas (4)–(5):

$$u = (\mathcal{Q} \otimes I_m)diag_{i=1}^n \bar{P}_s(s)(\mathcal{Q}^T \otimes I_m)y, \tag{4}$$

$$y = diag_{i=1}^n P(s)u, \tag{5}$$

where $\otimes$ is the Kronecker product, $u$ and $y$ represent the input and output of each agent, $n$ and $m$ highlight the number of agents and the maximum dimensional inputs/outputs of each agent, and $I_m$ illustrates an $m \times m$ identity matrix.

For a distributed consensus scheme, Eq. (4) is the same as Eq. (6) and as follows:

$$u_i = diag_{i=1}^n a_{in} P_{s,j}(s)(y_i - y_n), \tag{6}$$

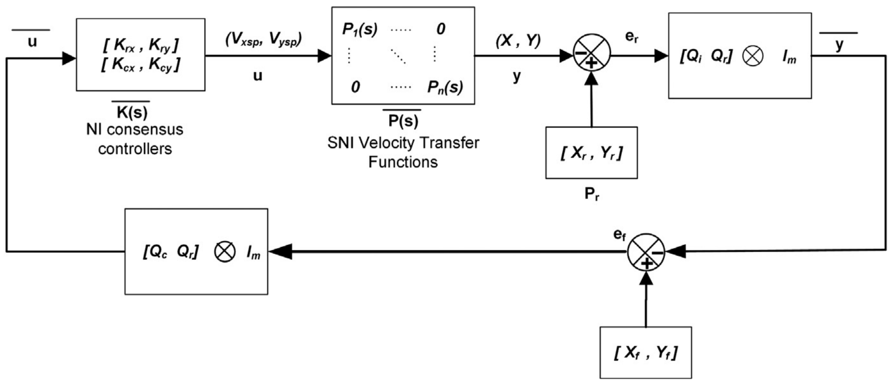



**Fig. 2.** Multi-UAV formation control architecture based on the NI systems theory.

where $a_{in}$ is the elements in the adjacency matrix and $j$ is the edge that connects agent $i$ to agent $n$.

The output feedback consensus between agents is achieved if and only if the following condition is satisfied:

$$\lim_{t\to\infty} \sum_{i=1}^{n} (y_i - y_{ss}) = 0, \tag{7}$$

where $y_{ss}$ is the final rendezvous point.

In order to generate a desired formation, new reference matrix, consensus matrix and the offset distance terms between UAVs are integrated to the above original consensus structure (see Fig. 2). The major experimental results show that the formation of UAVs is guaranteed and preserved during their rectangular movements. The overall Eqs. (8)–(13) of this architecture are given as follows:

$$P_r = 1_n \otimes r, \tag{8}$$

$$e_r = P_r - y, \tag{9}$$

$$y = diag_{i=1}^{n} P_i(s)u = \overline{P}(s)u \tag{10}$$

$$\bar{y} = (\mathcal{Q} \otimes I_m)e_r = ([\mathcal{Q}_a \ \mathcal{Q}_r] \otimes I_m)e_r, \tag{11}$$

$$e_f = P_f - \bar{y}, \tag{12}$$

$$u = (\mathcal{Q}^T \otimes I_m) diag_{i=1}^{n} K_i(s) e_f = ([\mathcal{Q}_b \ \mathcal{Q}_r] \otimes I_m)\overline{K}(s)e_f, \tag{13}$$

where $l$ illustrates the number of edges in information graph and $n$ highlights the number of agents. $I_m \in \mathbb{R}(m \times m)$ represents an $m - by - m$ identity matrix. $r \in \mathbb{R}(2\times \in \mathbb{R}(2 \times 1))$ denotes the reference position on the plane of the master, $P_r \in \mathbb{R}(2n \times 1)$ is the reference matrix, and $P_f$ corresponds to the relative position between the slave UAVs and the master UAV in the configured formation. $u \in \mathbb{R}(m \times 1)$ is the velocity set point input of each UAV on the $x$ and $y$ axes while the output $y \in \mathbb{R}(m \times 1)$ is the current position of each UAV. $\overline{u} \in \mathbb{R}(lm \times 1)$ and $\overline{y} \in \mathbb{R}(lm \times 1)$ are the input and output of overall network plant. $e_r$ is the error between the desired position of the master and its current one, while $e_f$ is the error between the desired relative position and the current one among the UAVs. $K(s) \in \mathbb{R}(lm \times lm)$ is the group of NI consensus and tracking controllers for the multiple UAVs. $\mathcal{Q}_a$, $\mathcal{Q}_b$ and $\mathcal{Q}_r$ are the incidence, consensus and reference matrix of agents respectively.

To verify the convergence of the NI consensus-based formation control method demonstrated in our previous work (Tran et al., 2017), a direct application of NI cooperative tracking, robust rendezvous is shown in a 3D scenario. We use UAV plants illustrated in Section 3 with the initial positions in 3 axis being [110; −90;0], [54; −90;0], [129;10;0] respectively. The relative position between agents $P_f$ = [0;0;0] and the pre-defined rendezvous point $P_r$ is [−60;40;70]. All SNI consensus controllers

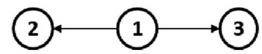



**Fig. 3.** Communication graph for three drones.

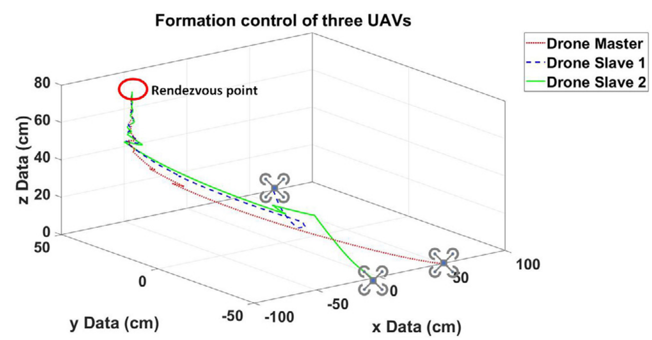



**Fig. 4.** 3D robust rendezvous of three SNI UAV systems via the original NI consensus protocol.

along three directions are chosen as $[\frac{-150}{0.5s+1}; \frac{-135}{0.5s+1}; \frac{-5.5}{0.5s+1}]$. The graph is exactly the same as Fig. 3 and the reference only sends information to the drone master, which gives $\mathcal{Q}_a = [1\ 1; -1\ 0; 0\ -1]^T$, $\mathcal{Q}_r = [1;0;0]^T$, and $\mathcal{Q}_b = [0\ 0; -1\ 0; -1\ 0]^T$. It can be seen from Fig. 4 that robust rendezvous is then guaranteed via the NI cooperative tracking controller. Three drones are properly driven from the initial positions to the final rendezvous point.

## 3. SNI/NI dynamic modeling for the UAV/UGV

To validate the efficacy and stability of the proposed control strategy, we derive the proper mathematical models to fully describe the UAV/UGV control systems. Firstly, the stability of UAV/UGV's velocity on the $x$–$y$ plane is ensured by a given PID controller. Its relevant parameters are appropriately tuned to meet the SNI system requirement. In addition, the roll/pitch angle set point of the UAV is limited at a maximum value of ±14 degrees to guarantee accurate representation by linear transfer functions. After the velocity controller has been determined and applied to the UAV/UGV systems, each robot's output control signals are changed from a lower to higher frequency to collect the overall dynamic behavior of our UAV and UGVs, including the absolute position, velocity, acceleration and Euler angles. Based on the data collected, we derive the closed-loop models via the ARMAX model system identification technique and then validate the fitting percentage between identified model output and real measured output (Eq. (14) and (15)) (Liu, Egan, & Santoso, 2015) before investigating their SNI/NI properties. In Ferrante,

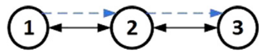



**Fig. 5.** Graph for the three robots.

Lanzon, and Ntogramatzidis (2016), it is concluded that the systems' SNI/NI properties remain unchanged to the additive model uncertainty. Hence, they will be guaranteed throughout the experiments. The general formula of this model in the z domain is given as:

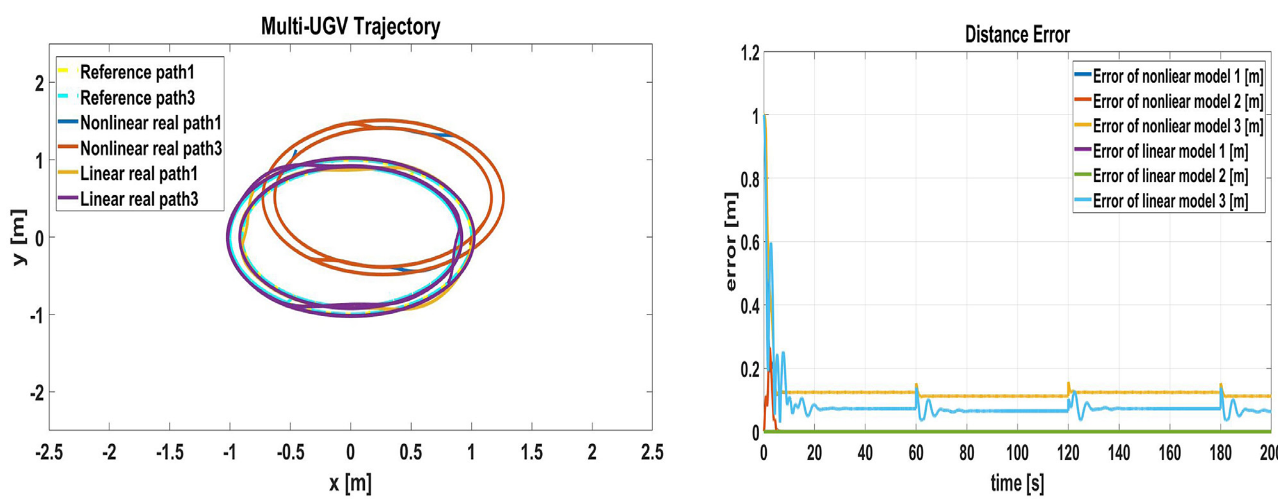
$$A(z)y(k) = B(z)u(k-d) + e(k), \tag{14}$$

$$fit\ (\%) = \left(1 - \frac{\|y - \tilde{y}\|}{\|y - \text{mean}(y)\|}\right) * 100\%, \tag{15}$$

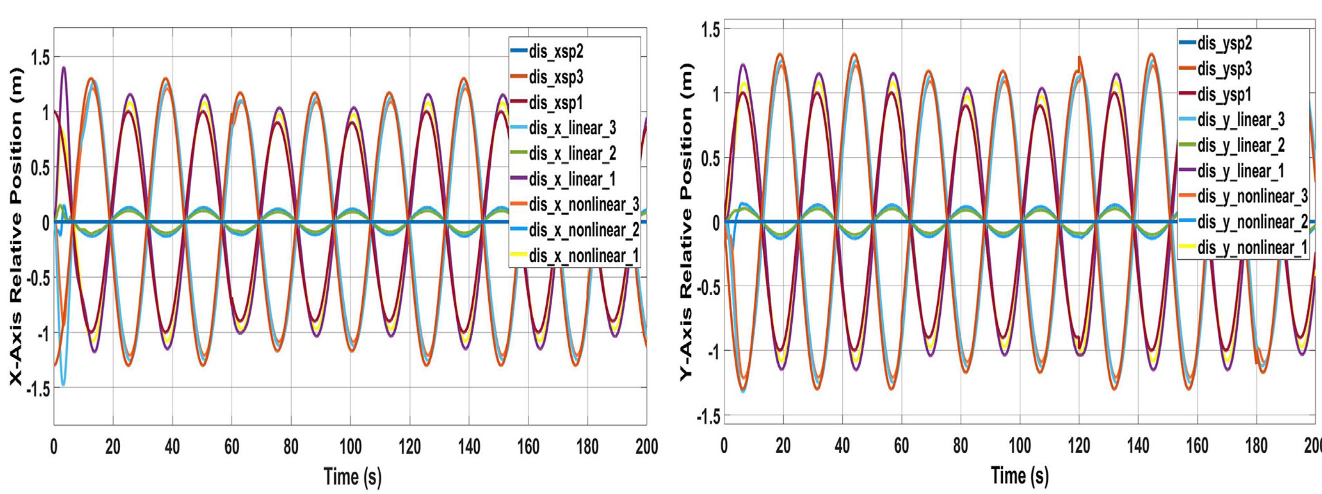
where $u(k)$ is the system input, $y(k)$ is the system output, $d$ is the system delay, $k$ is the sampling time and $e(k)$ is the error of disturbance in the system. $y$ and $\tilde{y}$ are the validation data output and the output of our real system respectively. $fit(.)$ is the percentage fit of the estimated model to the measured data.

Besides using the goodness-of-fit test to assess how well the model of behavior represents actual behavior towards mobile robots, one must also consider the ability of performing target-tracking and obstacle-avoidance tasks for the UGV SNI linear transfer functions relative to that of the UGV nonlinear kinematic models (e.g., in Martins, Sarcinelli-Filho, & Carelli, 2017). The two quadrotor transfer functions along the $x$ and $y$ axes, which are under the control of our NI formation technique presented in Tran et al. (2017), attempt to track two time-varying formations in simulation: circular-shaped references and eight-shaped references. The communication topology is provided in Fig. 5, where the master robot 1 shares its location to the slave robot 2 and the slave robot 2 shares its location to the slave robot 3. The maximum linear velocity is 20 cm/s, and the maximum angular rate is 10 degrees/s.

- Formation 1: The circular-shaped relative positions of the three robots were chosen as $[disx_1, disy_1] = [1+\cos(0.25t), -2.5+\sin(0.25t)]$, $[disx_2, disy_2] = [0, 0]$ and $[disx_3, disy_3] = [-1-\cos(0.25t), 2.5-\sin(0.25t)]$, in which $disx_i$ and $disy_i$ represent the desired relative positions of the robot $i$ over the $X-$ and $Y-$ axes. Both models are driven by the SNI consensus controllers with the general form of $\frac{-5}{0.01s+1}$. The results are interesting. It is worth noting that regarding the distance error values, only slight differences can be observed

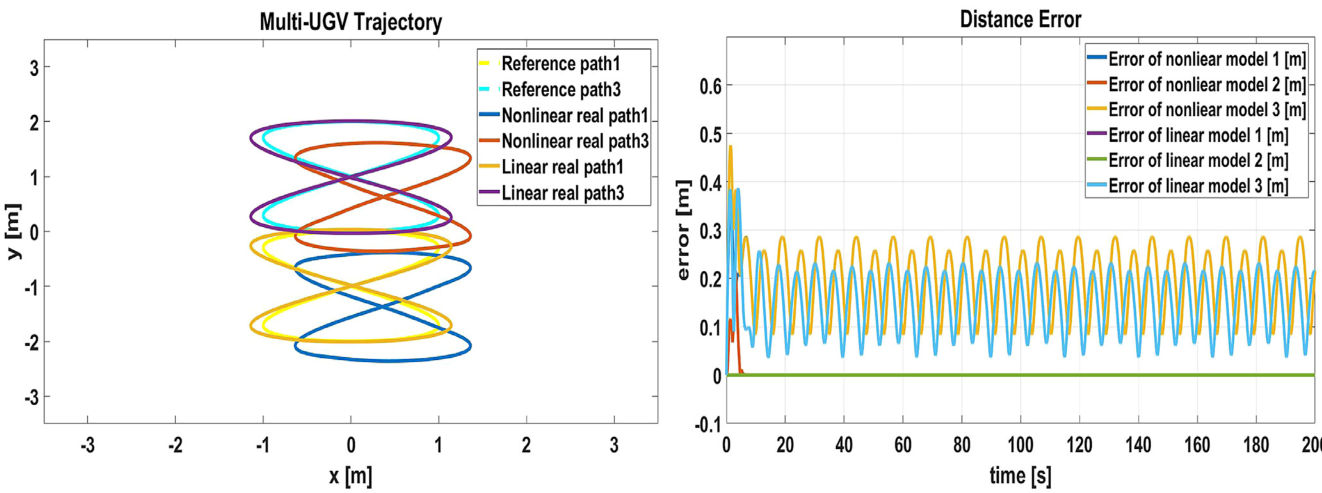



(a) *(left) Robot paths and (right) Distance errors.*

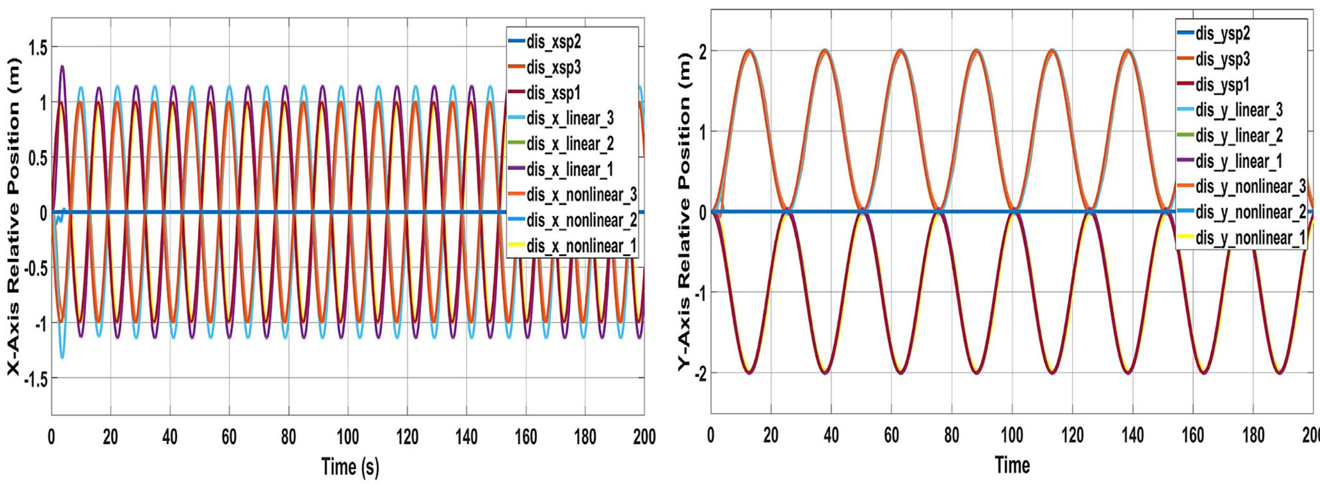



(b) *(left) X-axis and (right) Y-axis desired and actual relative positions.*

**Fig. 6.** SNI circular-shaped formation-control protocol and UGV ARMAX vs. kinematic models.




(a) *(left) Robot paths and (right) Distance errors.*




(b) *(left) X-axis and (right) Y-axis desired and actual relative positions.*

**Fig. 7.** SNI eight-shaped formation control protocol and UGV ARMAX vs. kinematic models.

**Table 1**
The UGV transfer functions between velocity x/velocity y set point (*velxsp*/*velysp*) and actual position x/position y output (*posx*/*posy*).

| | Transfer function |
|---|---|
| Velx | $posx/velxsp = \frac{-0.044s^3+5.16s^2-1860.11s+54489.05}{s^4+86.15s^3+54490.53s^2+29510.63s+1481.6}$ |
| Vely | $posy/velysp = \frac{-0.044s^3+4.24s^2-1494.74s+48347.83}{s^4+88.9s^3+54883.91s^2+49242.03s+1266.25}$ |

at the consensus outputs of two system models, as shown in Fig. 6(a). This conclusion received strong support from the achievements in Fig. 6(b), where both figures on each direction are quite similar and track well the desired trajectory over the experimental duration. Also, note that the NI method is able to track the given continuous reference-input signal with satisfactory performance on both models. This validates the accuracy level of the linear models in imitating the real behavior of the UGVs when performing two predefined tasks. It also provides ideal UGV transfer functions for further testing.

- Formation 2: The figure-of-eight relative positions of three robots, including the leader UAV and the two follower UGVs are defined by $[disx_1, disy_1] = [\sin(0.5t), 1-\cos(0.25t)]$, $[disx_2, disy_2] = [0, 0]$ and $[disx_3, disy_3] = [-\sin(0.5t), -1+\cos(0.25t)]$. All three SNI consensus controllers are selected as $\frac{-5}{0.01s+1}$.
Fig. 7(a) and (b) show that the output response signals of the UGV ARMAX model coordinated by the SNI controllers are similar to the one generated by the kinematic model. This is denoted by slight differences in the distance error values.

Through two tests, it is evident that linear UAV and UGV models, along with the given interval constraints, can mimic the actions of nonlinear UGV models very closely. As a result, these UGV transfer functions can be used to design the cooperative control methods in the coming sections.

After these transfer functions are found as shown in Tables 1 and 2, the SNI/NI property of each transfer function can be checked by plotting the Nyquist diagrams. As shown in the four images on the right side of Fig. 8, both Nyquist figures are contained below the real axis for the positive frequency. Therefore, all transfer functions found, whose data are collected by experimental tests, are SNIs. Also, the four images on the left side state that our transfer functions demonstrate a reasonably good accuracy of around 99% between the modeling and real test data.

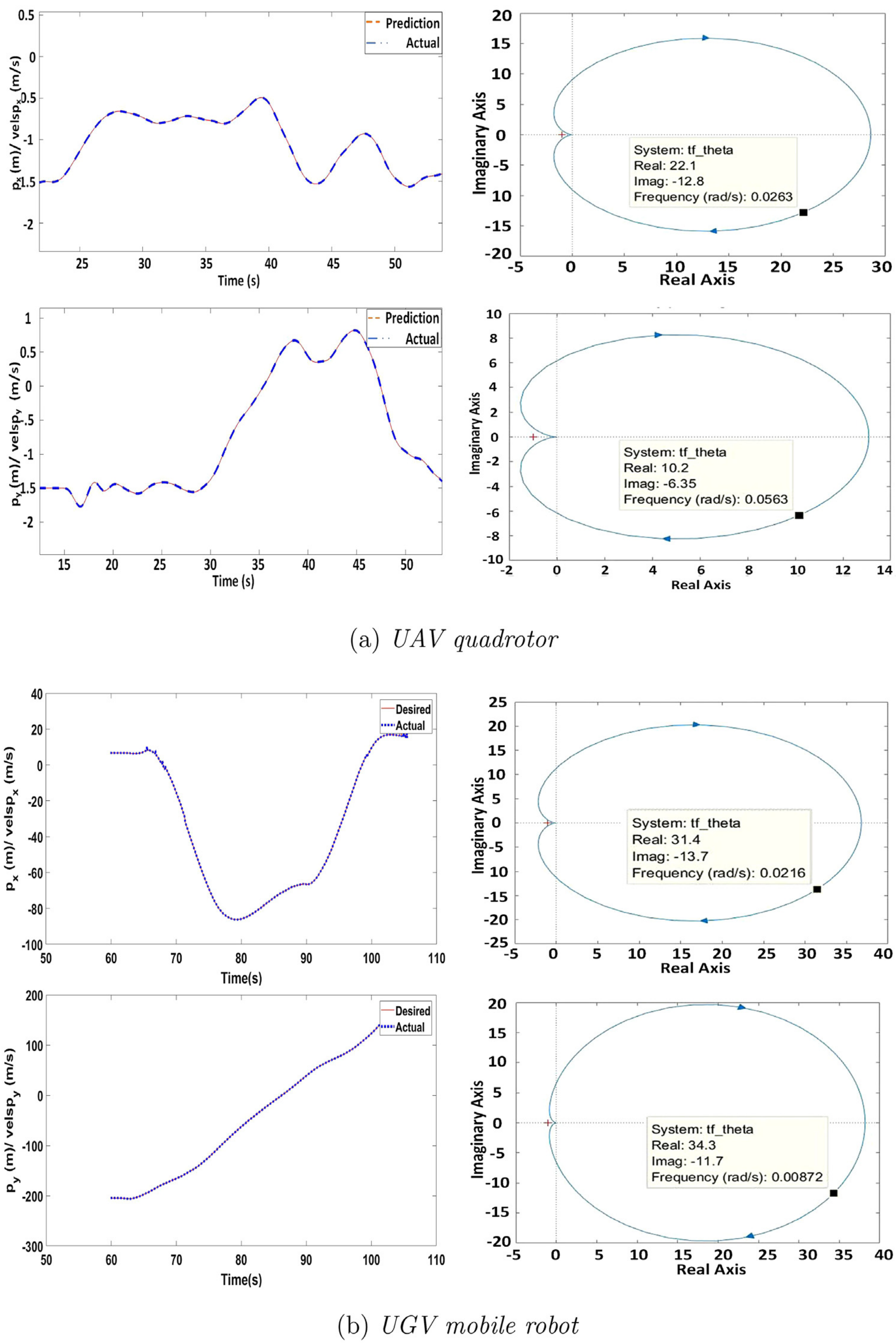



**Fig. 8.** (left) Accuracy level between modeling and the real data collected by identification experiments. (right) Nyquist plots of the UGV and UAVs original plants along $x$ and $y$ directions.

**Table 2**
The UAV transfer functions between velocity x/velocity y set point ($velxsp/velysp$) and actual position x/position y output ($posx/posy$).

| | Transfer function |
|---|---|
| Velx | posx/velxsp $= \frac{-0.0426s^3+6.76s^2-153.90s+33206.86}{s^4+86.15s^3+54490.53s^2+24730.20s+1161.87}$ |
| Vely | posy/velysp $= \frac{-0.05s^3-0.52s^2-1144.04s+29990.15}{s^4+101.06s^3+52978.06s^2+23472.45s+2291.96}$ |

## 4. Obstacle detection and avoidance control using NI systems theory

In this section, we would like to present a new way to detect obstacles using a collision detection vector ($\overrightarrow{ahead}$). In order to check for collision, the obstacles in this research are modeled as a circle-shaped object with a pre-defined safe radius $r$. A potential collision is found when $\overrightarrow{ahead}$ intersects with the obstacle circle. Meanwhile, an avoidance force, proportionally calculated based on the amount of overlap between ($\overrightarrow{ahead}$) and the circle, allows

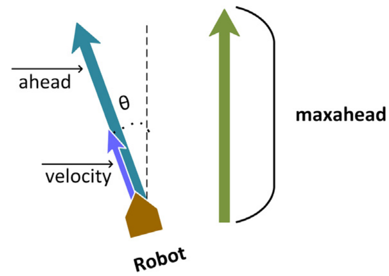



**Fig. 9.** The $\vec{ahead}$ heading vector based on the linear velocity of the robot.

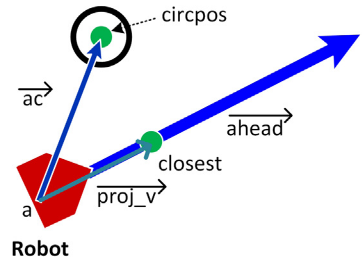



**Fig. 10.** The closest point on the ahead vector.

the robot to avoid the collision by steering around the constraints. This virtual force is then converted into a desired linear velocity on the x–y plane using Newton's second law. As soon as the virtual line-circle intersection ends, this force value becomes 0 and the relevant robot continues traveling on a consensus trajectory.

### 4.1. Obstacle detection

To achieve the purpose of precisely detecting obstacles in an uncertain environment, we base on the intersection between obstacle's virtual circle and the robot's heading vector.

Firstly, a vector aligned with the direction the robot's heading called $\overrightarrow{ahead}$ is defined. The function of the $\overrightarrow{ahead}$ is to predict a collision with obstacles. This vector's magnitude is equal to maximum distance of observation *max_ahead* multiplied by the ratio of the current speed to the maximum velocity that the vehicle can travel. This vector is fully described in (16)–(17) as follows:

$$\overrightarrow{ahead} = \frac{max_ahead\ V_r\ dynamic_length}{Vmax_r}, \tag{16}$$

$$dynamic_length = \frac{length(V_{ro})}{Vmax_{ro}}, \tag{17}$$

where $V_r$ and $V_{ro}$ (cm/s) denote the velocity of the robot and the relative velocity between the robot and the obstacle. *max_ahead (cm)* indicates how far the robot is able to detect obstacles. The larger the value of *max_ahead* is assigned, the earlier the robot will recognize the obstacle's position and start to avoid the obstacle. While $Vmax_r$ illustrates the maximum translational velocity during the robot movement, $Vmax_{ro}$ represents the maximum relative velocity. Once mobile obstacles move into the robot's field of view (FOV) range, the relative velocities between the robot and obstacle start being directly measured at every sample time, by means of a machine vision system installed on each robot. The relative velocity value measured at each sample time is immediately compared to the ones measured in the past and the maximum value will be sorted out and recorded. Any velocity value measured which is larger than $Vmax_{ro}$ will be recorded as $Vmax_{ro}$, otherwise it will be discarded. The comparison procedure is repeated until a potential collision between the robot and the obstacle is detected ($\overrightarrow{ahead}$ intersects the obstacle circle), then the latest value of $Vmax_{ro}$ will be utilized as an element to formulate the *ahead* robot vector. The variable *dynamic_length* will range from 0 to 1 (see Fig. 9).

Based on the $\overrightarrow{ahead}$ vector and the circle center coordinate, two significant values, including $\vec{ac}$ vector and *closest* point are computed as shown in Fig. 10. The actual $\vec{proj\_v}$ vector is found by multiplying its length and the ahead unit vector together. Next, the *closest* point is converted into global coordinates. The below formulas from Eq. (18) to Eq. (21) summarize the main steps to look for this point

$$\vec{ac} = circpos - a, \tag{18}$$

$$\left|\vec{proj\_v}\right| = \vec{ac}(\overrightarrow{ahead}/\left|\overrightarrow{ahead}\right|), \tag{19}$$

$$\vec{proj\_v} = \left|\vec{proj\_v}\right|(\overrightarrow{ahead}/\left|\overrightarrow{ahead}\right|), \tag{20}$$

$$closest = a + \vec{proj\_v}, \tag{21}$$

where *circpos* denotes the position of circle center. *a* is the starting point of the $\overrightarrow{ahead}$ vector. $\vec{proj\_v}$ is the projective vector of the $\vec{ac}$ vector onto the $\overrightarrow{ahead}$ vector.

Finally, a possible solution for collision detection is to inspect if the circle and the $\overrightarrow{ahead}$ vector intersect. To do this, we first determine the $\vec{av}$ vector from *closest* to *circpos*.

$$\vec{av} = closest - circpos. \tag{22}$$

If the length of this vector, *d*, is less than or equal to the circle's radius *r*, the circle and the $\overrightarrow{ahead}$ are intersecting. It means that there is a potential collision found (Fig. 11).

$$d <= r : \text{a collision is predicted}, \tag{23}$$

$$d > r : \text{no collision is predicted}. \tag{24}$$

However, not all obstacles in an unknown environment were analyzed the collision with the robot. Only the nearest one (based on the shortest distance from the obstacles to the robot) within the unsafe region, which is considered to be the most threatening, is selected for evaluation in the corresponding robot (Fig. 12).

$$\begin{aligned} Most_Thread &= \min(dis(robot_i))\forall i \in \{1..n\}, \\ & robot_i \in reg_i, \end{aligned} \tag{25}$$

where $dis_{robot_i}$ indicates the measured distance between the robot and the obstacles. $reg_i$ denotes the unsafe region containing the obstacles in it. *n* represents the number of obstacles whose regions intersect with each other.

### 4.2. Obstacle avoidance

A crucial parameter in collision response is the amount of overlap between the line vector $\overrightarrow{ahead}$ and the obstacle circle. This is represented as an $\vec{overlap}$ vector that has a similar direction as $\vec{av}$ whose length equals to the length deviation between the circle radius and the $\vec{av}$.

$$\vec{overlap} = (r - d)\vec{av}/d. \tag{26}$$

Our obstacle collision avoidance system is modeled as a spring–mass system where the vehicle is treated as a mass which is attached to a virtual spring. Based on Hooke's law, the potential avoidance force vector ($\vec{F_{avoid}}$) is defined as below:

$$\vec{F_{avoid}} = k\vec{overlap}, \tag{27}$$

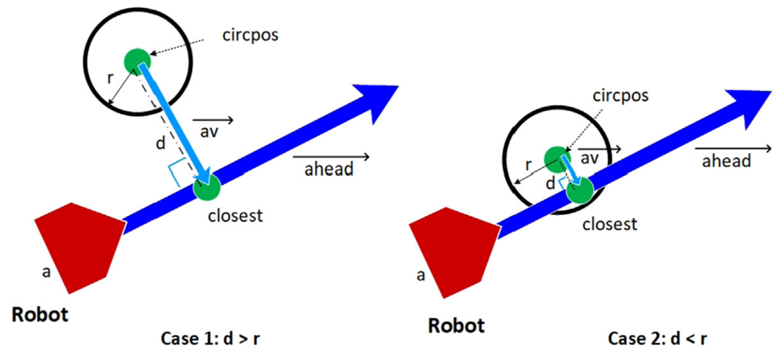



**Fig. 11.** Check for collision between the robot and the obstacle.

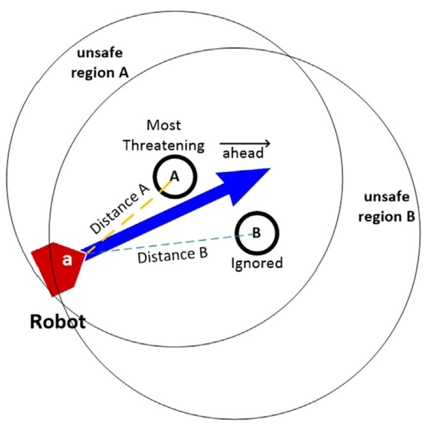



**Fig. 12.** The closest obstacle is selected based on the shortest distance.

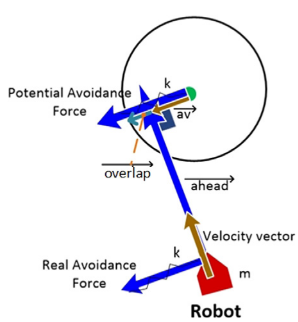



**Fig. 13.** The obstacle avoidance position set point based on the obstacle avoidance force.

where $k$ represents the stiffness of the spring used to generate the force.

The larger the overlap area is, the greater the avoidance force will be generated. The direction and magnitude of this force are then directly provided to the center of the robot. Finally, the process to compute the avoidance force is graphically depicted as in Fig. 13.

After the avoidance force and the mass of vehicle are known, the acceleration set point for obstacle avoidance is determined according to the Newton's second law of motion.

$$\vec{a_{asp}} = \vec{F_{avoid}}/m, \tag{28}$$

$$\vec{a_{asp}} = k\vec{overlap}/m, \tag{29}$$

where $m$ is the mass of vehicle and $a_{asp}$ is the desired acceleration on x–y axis.

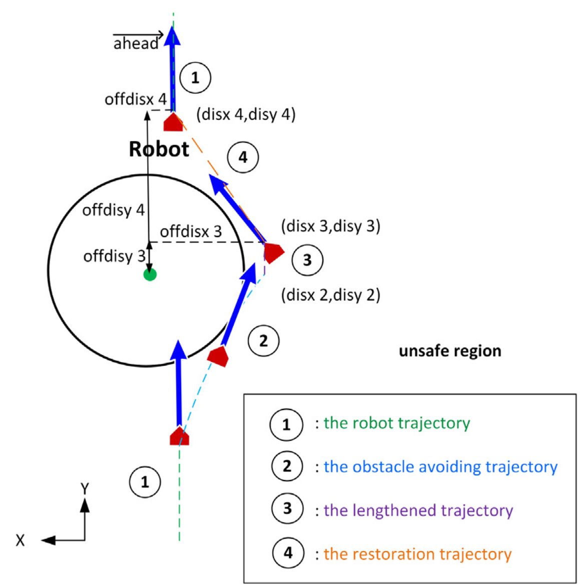



**Fig. 14.** The robot trajectory planning.

As previously explained in Tran et al. (2017), the consensus velocity on the x–y plane for each vehicle is produced by the NI consensus protocol to achieve the desired formation. For an NI uniform architecture, the obstacle avoidance transfer function must be converted into the Laplace domain as follows:

$$\overline{A(s)} = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & 0 \\ 0 & 0 & \cdots & a_N \end{bmatrix}, \tag{30}$$

with

$$a_i(s) = \frac{v_{asp_i}(s)}{\vec{overlap_i}} = \frac{k_i}{m_i s}, \forall i \in [1; N], \tag{31}$$

where $v_{asp}$ is the velocity set point of avoidance, $N$ is the number of robots, and $s$ is the Laplace domain.

Finally, the main results of Section 2.2 in Wang et al. (2015) show that the free body dynamics whose poles are at the origin and $a_i(\infty) = 0$ are NI plants. This then implies that the transfer function of the obstacle avoidance velocity in Eq. (30) is an NI plant containing a single integrator (a pole at the origin).

### 4.3. Restoration trajectory planning

It is mentioned in the previous section that as soon as the obstacle blocks the vehicle's traveling way, the avoiding velocity on the $x--y$ axis will be generated at the center of the vehicle to push itself away from the obstacle. This action leads to reconstruction in the trajectory planner. The robot, which is traveling along its trajectory, will react against the additional avoidance velocity by planning a new route to pass the obstacle, named obstacle avoiding trajectory. This second trajectory will be produced by the robot until non-intersection or non-collision is satisfied and the repulsive force is vanished. At this time, a lengthened trajectory is required to ensure there will be no further touch between that robot and the obstacle circle. The last destination of the second trajectory on the $x$ axis (*disx 2*) is also maintained in the third one (*disx 3*) while an offset distance of *offdisy 3* from the coordinate of obstacle's center is assigned to the robot coordinate on the y axis (*disy 3*). After reaching the terminal point of this trajectory, the robot is oriented towards the original path by moving along the restoration trajectory. The next desired position on the $x$ axis (*disx 4*) is restored to the old coordinate before the intersection occurs while that on the $y$ axis (*disy 4*) is increased by a more considerable amount than *offdisy 3*, called *offdisy 4*. If the ending point of the fourth route is too near to the final destination, our robots will select the final destination to approach. After the fourth trajectory is completed, the corresponding robot will continue to follow the robot trajectory. The above process will be repeated from the beginning as long as more obstacles are still existing along the robot's moving path. Fig. 14 sums up all steps of the safe-paths generation.

## 5. Formation-avoidance (FA) architecture based on the NI systems theory

As reviewed in Section 2.2, an NI consensus architecture for multi-UAVs is introduced and applied to the two UAVs (Tran et al., 2017). In the indoor experimental result shown in this paper, a master UAV tracks the rectangular trajectory to correct its behavior while a follower drives itself to maintain the relative position with respect to the absolute position of the master. The significant equations describing the consensus and formation control for two robots are given as follows:

$$\begin{bmatrix} V_{xtsp_1} \\ V_{ytsp_1} \\ V_{xtsp_2} \\ V_{ytsp_2} \end{bmatrix} = \begin{bmatrix} \mathrm{Kr}_x(\mathrm{X}_1 + \mathrm{X}_r) \\ \mathrm{Kr}_y(\mathrm{Y}_1 + \mathrm{Y}_r) \\ \mathrm{Kc}_x[\mathrm{X}_1 - \mathrm{X}_2 + \mathrm{X}_f] \\ \mathrm{Kc}_y[\mathrm{Y}_1 - \mathrm{Y}_2 + \mathrm{Y}_f] \end{bmatrix} \tag{32}$$

$$= \begin{bmatrix} \mathrm{Kr}_x(\mathrm{X}_1 + \mathrm{X}_r) \\ \mathrm{Kr}_y(\mathrm{Y}_1 + \mathrm{Y}_r) \\ \mathrm{Kc}_x[\mathrm{d}_x + \mathrm{X}_f] \\ \mathrm{Kc}_y[\mathrm{d}_y + \mathrm{Y}_f] \end{bmatrix}, \tag{33}$$

where $Kr_x$, $Kr_y$ and $Kc_x$, $Kc_y$ are the reference and consensus gain values respectively for the NI velocity controllers on the two coordinates $x$ and $y$. $(V_{xtsp1}, V_{ytsp1})$, $(X_1, Y_1)$ and $(V_{xtsp2}, V_{ytsp2})$, $(X_2, Y_2)$ define the desired velocity input and the absolute position output of the leader/follower on the horizontal plane, respectively. The variables $d_x$ and $d_y$ denote the relative distances between the leader and its followers along the $x$ and $y$ axes. $(X_r, Y_r)$ and $(X_f, Y_f)$ represent the corresponding pairs of the desired trajectory references of the leader and the expected relative distances from the follower to the leader on the $x$ and $y$ axes.

For general situation, given multiple SNI plants with $N > 1$ robots, the formation-avoidance velocity references are expressed as below:

$$u = \begin{bmatrix} V^{(t)}_{xsp_L} \\ V^{(t)}_{ysp_L} \\ V^{(t)}_{xspF_1} \\ V^{(t)}_{yspF_1} \\ \vdots \\ V^{(t)}_{xspF_{N-1}} \\ V^{(t)}_{yspF_{N-1}} \end{bmatrix} = \begin{bmatrix} Kr_x(X_{L_x} + X_r) \\ Kr_y(Y_{L_y} + Y_r) \\ Kc_{x_1}(d_{F_{x_1}} + X_{F_1}) \\ Kc_{y_1}(d_{F_{y_1}} + Y_{F_1}) \\ \vdots \\ Kc_{x_{N-1}}(d_{F_{x_{N-1}}} + X_{F_{N-1}}) \\ Kc_{y_{N-1}}(d_{F_{y_{N-1}}} + Y_{F_{N-1}}) \end{bmatrix}, \tag{34}$$

$$u = \begin{bmatrix} V^{(t)}_{xsp_L} \\ V^{(t)}_{ysp_L} \\ V^{(t)}_{xspF_1} \\ V^{(t)}_{yspF_1} \\ \vdots \\ V^{(t)}_{xspF_{N-1}} \\ V^{(t)}_{yspF_{N-1}} \end{bmatrix} = \begin{bmatrix} Kr_x \Delta x_{L_r} \\ Kr_y \Delta y_{L_r} \\ Kc_{x_1} \Delta x_{LF_1} \\ Kc_{y_1} \Delta y_{LF_1} \\ \vdots \\ Kc_{x_{N-1}} \Delta x_{LF_{N-1}} \\ Kc_{y_{N-1}} \Delta y_{LF_{N-1}} \end{bmatrix}, \tag{35}$$

$$y = \begin{bmatrix} X^{(t)}_L \\ Y^{(t)}_L \\ X^{(t)}_{F_1} \\ Y^{(t)}_{F_1} \\ \vdots \\ X^{(t)}_{F_{N-1}} \\ Y^{(t)}_{F_{N-1}} \end{bmatrix} = \overline{P(s)}u, \tag{36}$$

where $(X_{L_x}, Y_{L_x})$ and $(X_r, Y_r)$ are the actual and desired position on the $x$ and $y$ directions while $(X_{F_i}, Y_{F_i})$ and $(d_{F_{x_1}}, d_{F_{y_1}})$ are the actual and desired relative position between L and F. $(Kr_x, Kr_y)$, $(Kc_x, Kc_y)$ are the SNI/NI consensus controllers of L and F.

Now, an appropriate cooperative strategy must be designed to connect three fundamental tasks: maintaining a given formation, approaching the destination, and guaranteeing collision avoidance. Based on Lemma 2.1, NI obstacle avoidance transfer functions $\underline{A}(s)$, NI consensus controllers $K(s)$ and SNI vehicle dynamics $\overline{P(s)}$ can work and interact effectively with each other by positive connections, namely, priority weights for each task $[a_x; a_y]$. The incorporation of the two transfer functions is exactly described as in (37).

$$\begin{bmatrix} V^{(t)}_{xsp_L} \\ V^{(t)}_{ysp_L} \\ V^{(t)}_{xspF_i} \\ V^{(t)}_{yspF_i} \end{bmatrix} = \begin{bmatrix} Kr_x[a_{x_{0,1}} \Delta x_{L_r} + a_{x_{0,2}} vx_{a_{sp_0}}] \\ Kr_y[a_{y_{0,1}} \Delta y_{L_r} + a_{y_{0,2}} vy_{a_{sp_0}}] \\ Kc_{x_i}[a_{x_{i,1}} \Delta x_{LF_i} + a_{x_{i,2}} vx_{a_{sp_i}}] \\ Kc_{y_i}[a_{y_{i,1}} \Delta y_{LF_i} + a_{y_{i,2}} vy_{a_{sp_i}}] \end{bmatrix}, \tag{37}$$

where $(a_{x_{0,1}}, a_{y_{0,1}})$, are the task priority weights of the leader while $(a_{x_{i,1}}, a_{y_{i,1}})$, $\forall i = 1, \ldots, N-1$, are the task priority weights of the $i$th slave, respectively. $(vx_{a_{sp_0}}, vy_{a_{sp_0}}), \ldots, (vx_{a_{sp_i}}, vy_{a_{sp_i}})$ are the desired velocities of the leader and the followers on the $x$ and $y$ axes provided by the obstacle avoidance approach.

NI consensus is achieved by using a virtual mass–spring system connecting the agents so that they do not get too close or too far away from each other. The distributed NI obstacle avoidance scheme also has the same structure as the NI consensus where the obstacle is placed on top of an artificial hill which exerts a gravitational force to keep the agents away from it. With this new

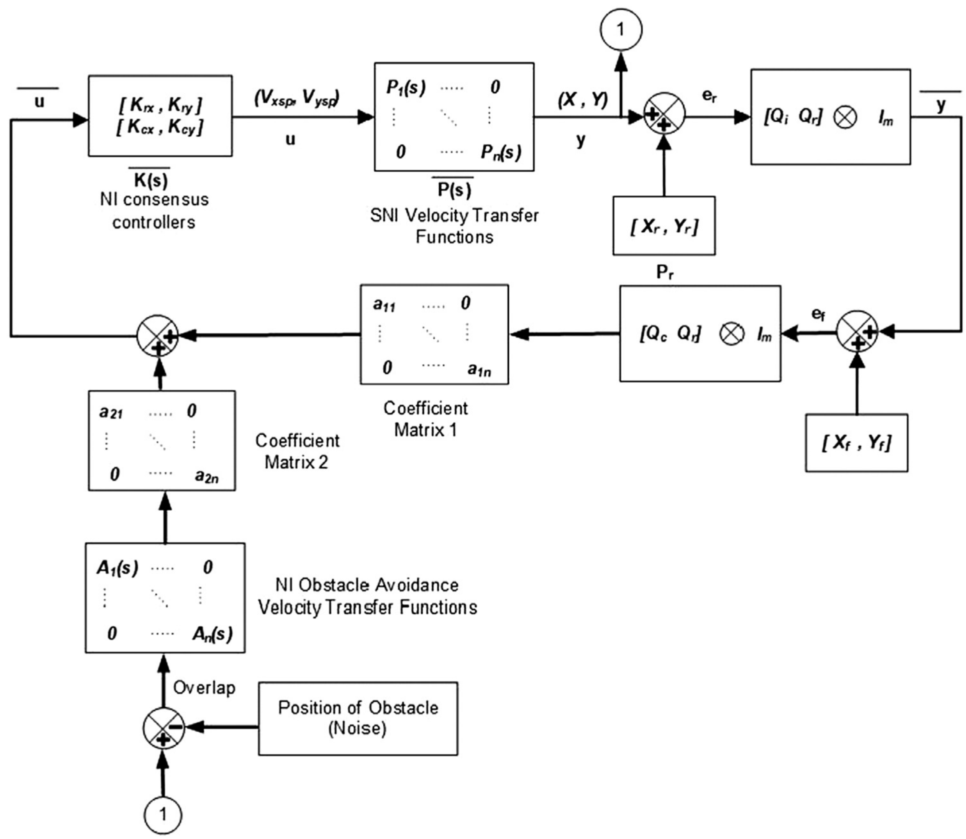



**Fig. 15.** The FA architecture using NI systems theory.

structure, once a potential collision is predicted, the NI consensus and NI avoidance spring work together to maintain the proper distance between agents. Additionally, the absolute position of each obstacle is considered as a noise in the FA architecture. Since the obstacle avoidance transfer functions are only NI plants, any oscillations affecting these systems cannot be reduced with time due to the lack of the damping coefficient in their structure. Nevertheless, by connecting these NI transfer functions to the SNI agent plants in the FA architecture, these unexpected disturbances may die out.

It is clear that in some previously published literature (Abeywickrama, Jayawickrama, He, & Dutkiewicz, 2017; Dai et al., 2015; Lee & Chwa, 2018), only formation control and collision avoidance problems were designed, whereas their homogeneity and synchronization have not fully been taken into account. By using (37), a homogeneous and synchronous consensus control protocol for all problems and an internal stability of $[\overline{K(s)}, \overline{P(s)}, \overline{A(s)}]$ are guaranteed via the NI framework.

After velocity setpoints are determined, the desired yaw and escape angle around the z axis of the UGVs are computed by *atan2* equation as follows:

$$\omega_{sp_i} = atan2(Y_{sp_i} - Y_i, X_{sp_i} - X_i), \tag{38}$$

Let $\omega_{sp_i}$ represent the yaw angle setpoints.

For more details, all calculation processes are summarized by a graphic diagram, as shown in Fig. 15.

## 6. The cooperative strategy based on the FA architecture

During switching process between various tasks, clear oscillations may be seen in the UGVs' real paths, which is a drawback of existing methods. Therefore, we present a novel cooperative strategy that allows each vehicle to steer away from the closest obstacle in its path and then to resume the consensus between robots until reaching the target point. This approach considers three possibilities of the collision: the potential crash is detected by only the master, by only the slaves and by both master–slaves. Afterwards, a set of actions corresponding to each case is implemented by the robots.

### 6.1. Case 1: The potential collision is found by only the master

In this situation, there is a risk of collision with the master. The master responds to this detection by generating a steering force while the followers are guided by the current desired relative distance on the vertical axis (the y axis) and the relative distance on the horizontal axis (the $x$ axis), measured by the camera as soon as the collision is predicted using Eq. (39). After the avoiding process from the master is accomplished, Eq. (37) returns to Eq. (33) (Fig. 16).

$$\begin{bmatrix} V^{(t)}_{xsp_L} \\ V^{(t)}_{ysp_L} \\ V^{(t)}_{xsp_{F_i}} \\ V^{(t)}_{ysp_{F_i}} \end{bmatrix} = \begin{bmatrix} Kr_x(X_{L_x} + X_r) \\ Kr_y(Y_{L_y} + Y_r) \\ Kc_{x_i}(d_{F_{x_i}} + oldX_{F_i}) \\ Kc_{y_i}(d_{F_{y_i}} + curY_{F_i}) \end{bmatrix} \tag{39}$$

### 6.2. Case 2: The potential collision is found by only the follower

When the closest obstacle in the environment is obstructing the follower's movement, the follower will immediately send the data of relative distance between it and the relevant obstacle to the master. Upon receiving this data, the master will compute the obstacle coordinate based on the $Y_f$ relative distance between the master and the follower as well as the measured distance $dis_{folob}$ between the follower and the obstacle Eq. (40). Next, three

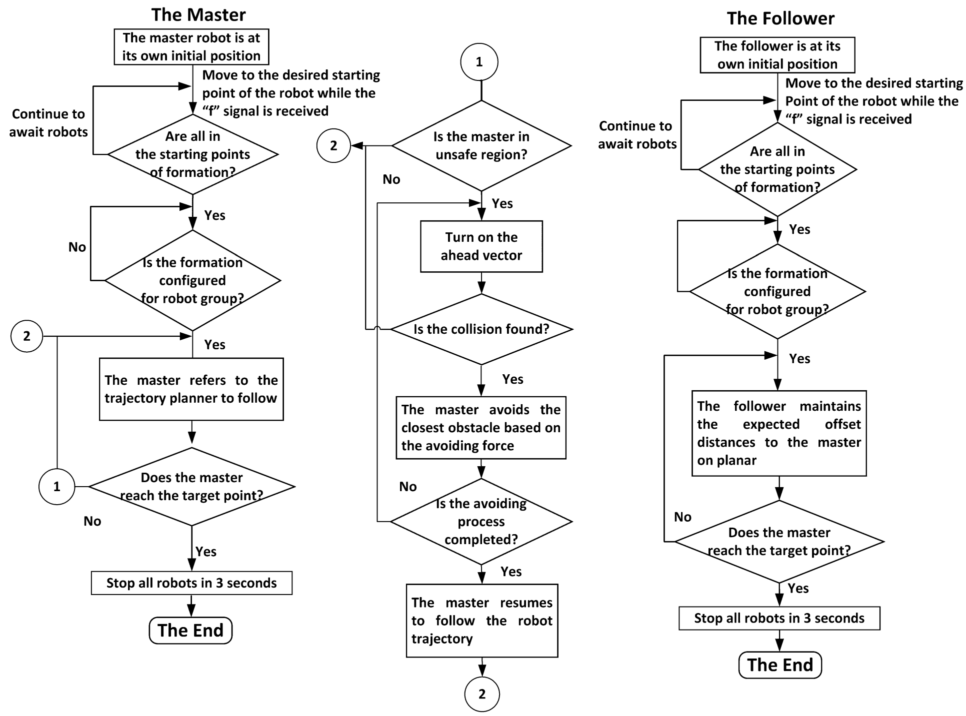



**Fig. 16.** The cooperative strategy of NI FA architecture in Case 1.

essential parameters on the vertical axis, comprising the location of the obstacle ($Y_{ob}$), *offdisy 4*, and the desired relative distance between the master and the follower ($Y_f$), are used to calculate the desired stopping point on the y axis where the master must wait Eqs. (41)–(43). If more than one slave is avoiding obstacles, the coordinate of the obstacle, which is closest to the target, will be used to calculate the temporary stopping point for the master. The master must wait at that position ($Xstop_r$, $Ystop_r$) until the follower approaches the final destination on its restoration trajectory. The wait performed by the master facilitates the position consensus and the pre-defined formation maintenance in next step. After that, two robots are driven to the target point by Eq. (33) (Fig. 17).

$$Y_{ob} = Y_{master} - Y_f + dis_{folob}, \tag{40}$$

$$Xstop_r = curX_r, \tag{41}$$

$$Ystop_r = Y_{ob} + offdisy4 + Y_f, \tag{42}$$

$$\begin{bmatrix} V^{(t)}_{xsp_L} \\ V^{(t)}_{ysp_L} \\ V^{(t)}_{xsp_{F_1}} \\ V^{(t)}_{ysp_{F_1}} \\ \vdots \\ V^{(t)}_{xsp_{F_{N-1}}} \\ V^{(t)}_{ysp_{F_{N-1}}} \end{bmatrix} = \begin{bmatrix} Kr_x(X_{L_x} + Xstop_r) \\ Kr_y(Y_{L_y} + Ystop_r) \\ Kc_{x_1}(d_{F_{x_1}} + X_{F_1}) \\ Kc_{y_1}(d_{F_{y_1}} + Y_{F_1}) \\ \vdots \\ Kc_{x_{N-1}}(d_{F_{x_{N-1}}} + X_{F_{N-1}}) \\ Kc_{y_{N-1}}(d_{F_{y_{N-1}}} + Y_{F_{N-1}}) \end{bmatrix} \tag{43}$$

### 6.3. Case 3: The potential collision is found by both robots

This issue is solved by a combination of the two approaches previously provided in Case 1 and 2. As soon as the two robots meet the condition for a collision simultaneously, the desired stopping point is calculated by the master (Case 2) and the alternative references are processed by the follower (Case 1).

## 7. Stability proof

In this section, we recall the three concepts for SNI/NI MIMO systems presented in Petersen and Lanzon (2010) and Wang et al. (2015).

**Lemma 7.1.** *$\overline{P}(s)$ is SNI if and only if each member $P_i(s)$ is SNI (Wang et al., 2015).*

**Lemma 7.2.** *Given any SNI MIMO system $\overline{P}(s)$ and NI/SNI MIMO controller $\overline{K}(s)$, we obtain a stability result for the whole structure as follows (Wang et al., 2015):*

$$\lambda_{max}(\overline{P}(0))\lambda_{max}(\overline{K}(0)) < \frac{1}{\lambda_{max}(\mathcal{Q}\mathcal{Q}^T)}, \tag{44}$$

$$\lambda_{max}(\overline{P}(0)) < \frac{1}{\lambda_{max}(\mathcal{Q}\mathcal{Q}^T)\ \lambda_{max}(\overline{K}(0))}. \tag{45}$$

As proven in Section 3, each velocity transfer function $P_i(s)$ representing the UAV or UGV systems has the SNI property; hence, their $\overline{P}(s)$ matrix is SNI according to Lemma 7.1.

As presented in Eq. (30), the velocity transfer functions of the obstacle avoidance problem are NI plants. Additionally, those of the formation control with its communication graph are SNI systems as shown in Tran et al. (2017). According to Eq. (37), the overall velocity transfer functions of the FA architecture on the x–y plane, giving a positive connection between the NI systems and the SNI ones, are SNI systems. For example, the $X - axis$ SNI velocity transfer function of UGV with its topology (*TF1*) and the NI avoidance transfer function of the UGV with $k$ gain of 0.225 (*TF2*) are used to prove this lemma. As shown in Fig. 18, its final figure (*TF*) demonstrates an SNI system (*M(s)*).

Since $\mathcal{Q}\mathcal{Q}^T$ have fixed values which are greater than 0, leading to $\lambda_{max}(\mathcal{Q}\mathcal{Q}^T) > 0$, only their NI controllers $\overline{K}(0)$ can be tuned to achieve the stability for the whole structure. For this reason, we select the negative gain values for $\overline{K}(s)$ so that the required stability is always guaranteed.

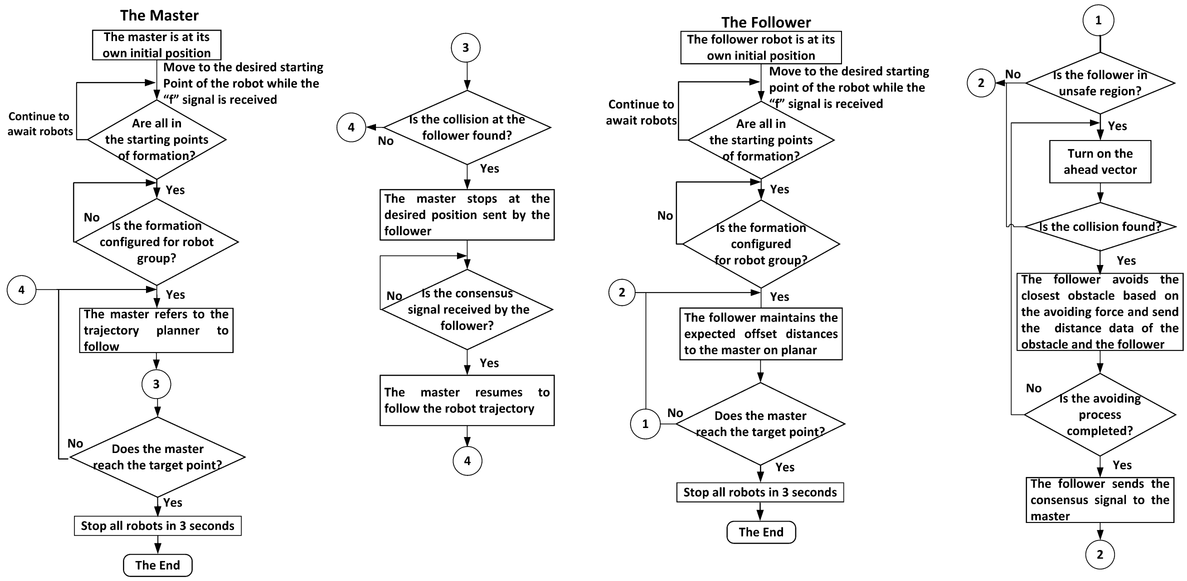



**Fig. 17.** The cooperative strategy of NI FA architecture in Case 2.

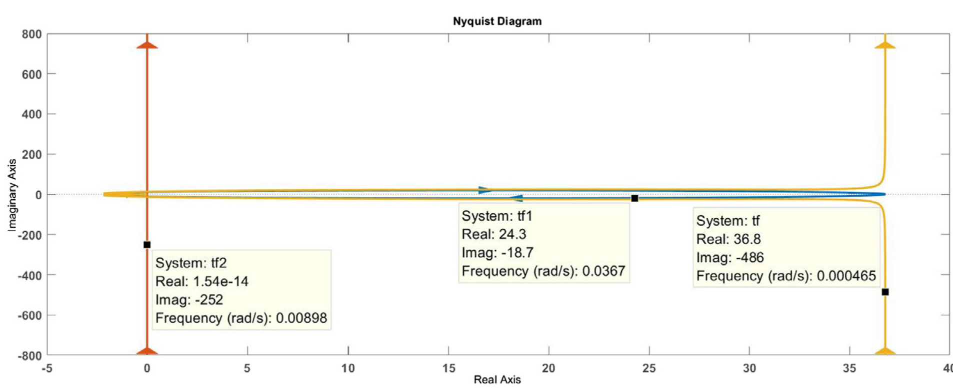



**Fig. 18.** The SNI property of a positive connection between an SNI system (TF1) and an NI system (TF2).

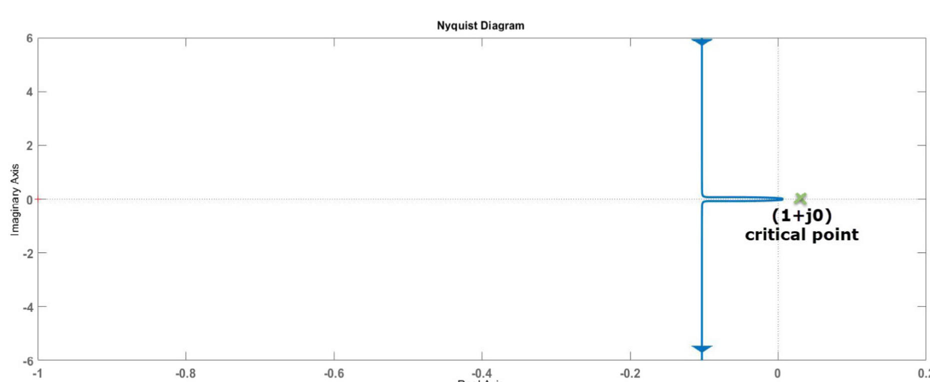



**Fig. 19.** The internal stability of the whole FA architecture.

Regarding the UGV's velocity transfer function, their real DC gains at the zero frequency are equal to $\frac{54489.05}{1481.6} = 36.77$ (Condition 1). In addition, $\mathcal{Q}\mathcal{Q}^T > 0$ (Condition 2). All NI controllers are chosen with the negative values ($\overline{K} < 0$) (Condition 3). Based on Conditions 1–3, the stability criteria in Lemma 7.2 are naturally satisfied. To strengthen our argument, the figure of the complete structure plotted in the Nyquist diagram never encircles the critical point (1;+j0) (Fig. 19).

## 8. Experimental apparatus

### 8.1. Hardware/software configuration

One UAV (F450 quadrotor platform), two heterogeneous UGVs (Pioneer P3-AT and P3-DX) and two obstacles (plastic packing boxes) are used in our experiments. The position and orientation of agents and obstacles are analyzed and measured by VMCS

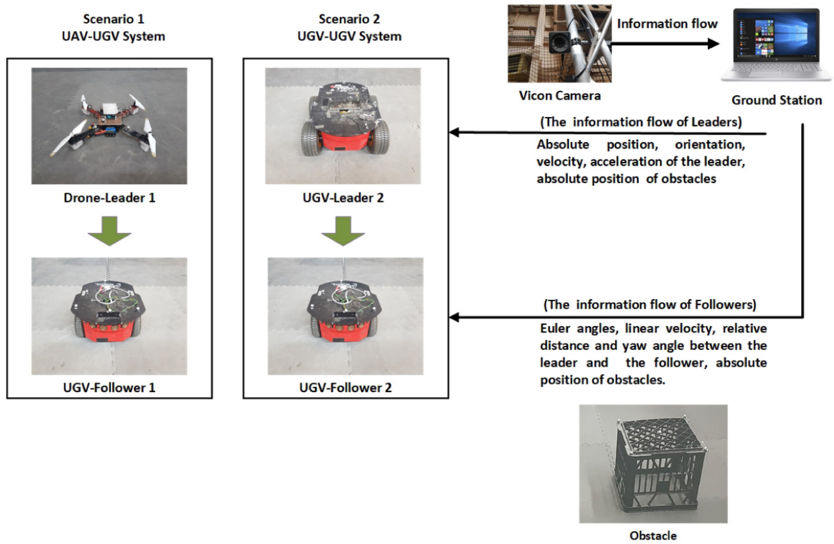



**Fig. 20.** Overall architecture diagram.

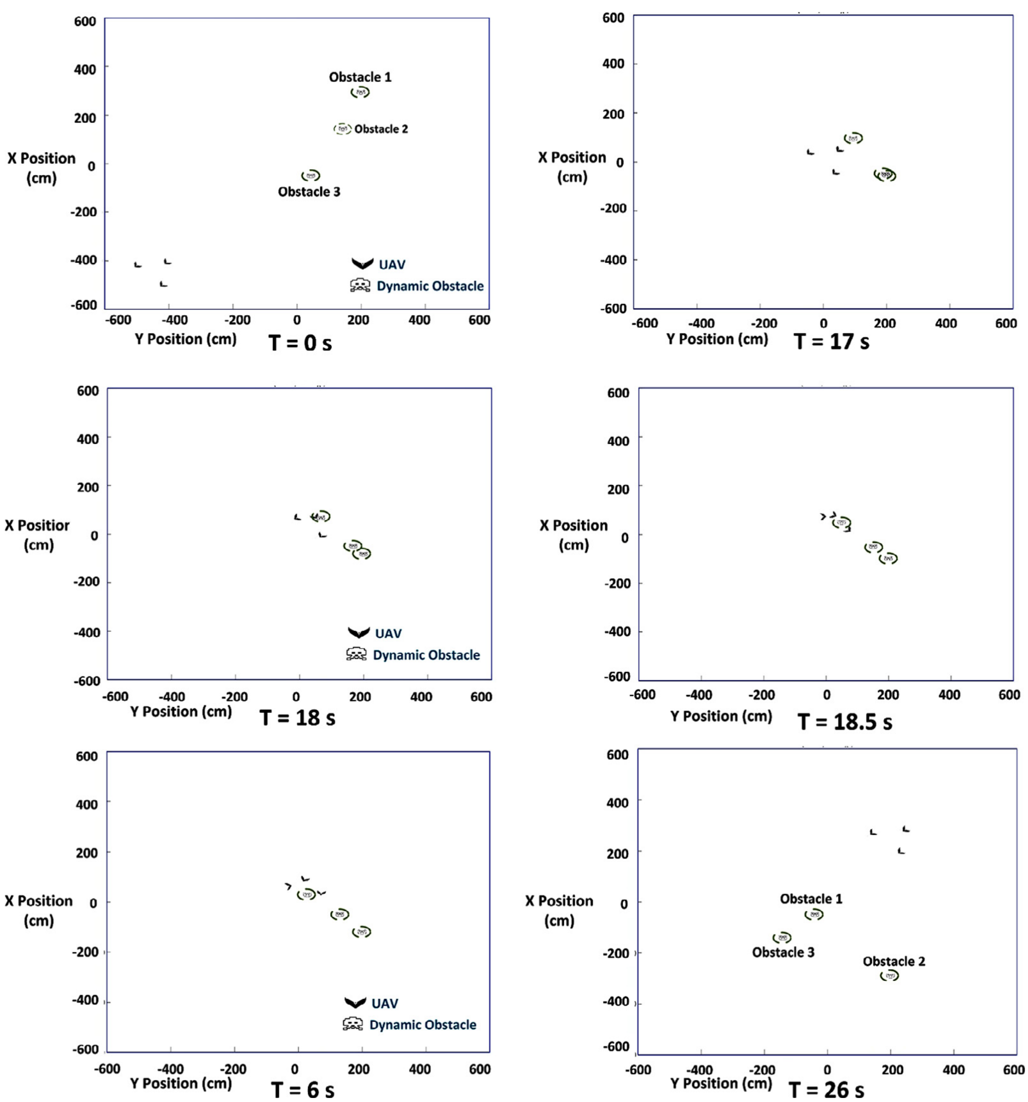



**Fig. 21.** Output trajectories of the three quadrotors using the NI-SFOA control law.

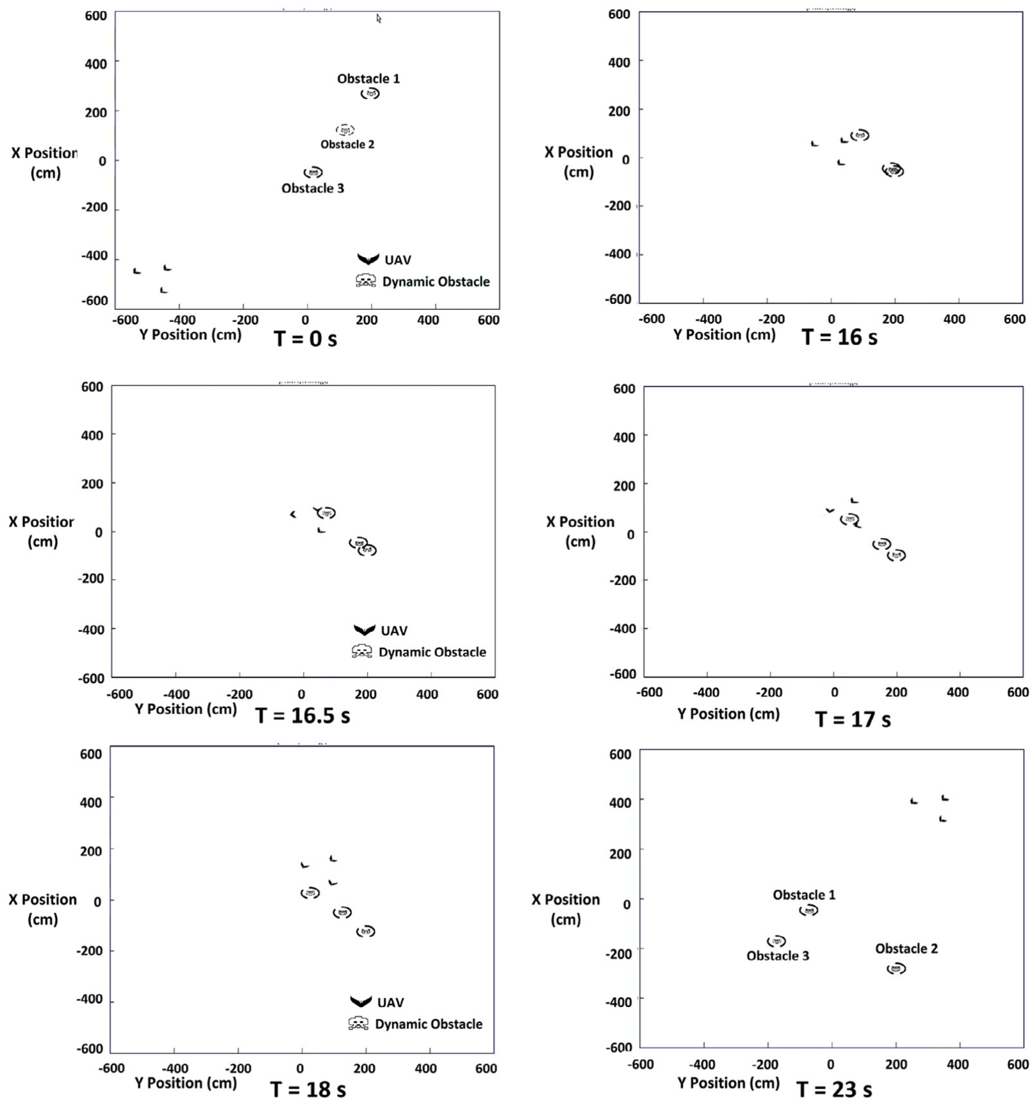



**Fig. 22.** Output trajectories of the three quadrotors using the proposed NI methodology.

which broadcasts this information continuously at a frequency of 100 Hz via a UDP network protocol. The interaction protocols between the robots and between the ground station and the robots are achieved using the Robot Operating System (ROS). On the UAV, an IMU sensor is installed to measure the roll and pitch motions during flight test while the wheels of each UGV are equipped with optical encoders to estimate the linear velocity, moving distance and yaw rate.

### *8.2. Overall architecture*

Our leader–follower (L–F) control network architecture is designed to perform cooperative scenarios involving both UAV–UGV and UGV–UGV system. The functionality and the task of each block are adequately described in Fig. 20.

## 9. Results

The altitude of the UAV is stabilized at 70 cm above the ground plane by a PID controller which maintains a clearance of 28 cm between the UAV and the top of UGVs. Simulations in MATLAB and experiments in an indoor environment were conducted to analyze and validate the feasibility of our proposed methodologies using both UAV and UGV platforms.

### *9.1. Simulation results*

Four cases are given to illustrate the obtained results of current implementations. The first case compares the dynamic obstacle avoidance capability of the proposed approach with that of the NI-SFOA. The second case is to show the NI formation control scheme of the UAV–UGV system involving the one UAV and the two UGVs, while the third case performs an inverted V-formation and the obstacle avoidance generated by the master UAV and the five follower UGVs. The final case estimates the efficacy of the proposed method with respect to the APF.

#### *9.1.1. Simulation 1*

In the first test, a flock of the three quadrotor UAVs under the two comparative methods is commanded to move to the goal point and evade the three mobile obstacles simultaneously. The two velocity transfer functions from the UAV on x–y plane (Table 2) are converted into proper state-space models before they are added in our script M-file. The purpose of this step is to simulate accurately the behavior of UAV throughout the obstacle avoiding process. The fundamental parameters are given as follows: $\overline{K} = (\mathrm{Kr}_x,\mathrm{Kr}_y) = (-0.7;\,-0.7)$, $\overline{A} = (\mathrm{kx;ky}) = (0.35;0.35)$, $m = 1$ kg, $max\_ahead = 10$ cm. The UAV's maximum translational

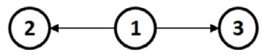



**Fig. 23.** Graph illustrates the communication between the three SNI systems.

speed, $V_r$, and the UAV's maximum avoiding speed, $Vmax_{ro}$, are 5 cm/s. Further, the radius of both obstacle circles, $r$, is 26 cm. The initial positions of the objects, involving UAV and three obstacles, are selected randomly in a 2D scenario whereas the expected destination for UAV is located at (450, 450) cm. The translational velocities on the $x$ and $y$ axes of the three obstacles are chosen as (1.5;1.5), (1.5;0) and (0;1.5) cm/s. The main parameters of the NI-SFOA algorithm are repeated as in the simulation experiments (Tran et al., 2020b).

As shown in Figs. 21 and 22, the collaborative quadrotors can generate the desired V-shaped formation within only 2 s before encountering the mobile obstacles at the 17th s. It is clear that the novel NI obstacle avoidance controller outperforms the NI-SFOA when reaching the target point quicker (less than 2 s), stabilizing faster back to the desired references, and producing the smoother avoiding routes. In contrast, the NI-SFOA yields unsatisfactory performance when the master robot hits one obstacle, and the slave quadrotor's flight path dramatically fluctuates (see Fig. 21). This is because, in our strategy, each robot's repulsive force is adapted to the intersection area. Moreover, the length of the $\overrightarrow{ahead}$ vector is adjusted based on the robot's velocity. Consequently, our NI method can operate well in various scenarios without taking much time to retune the control parameters like the NI-SFOA.

*9.1.2. Simulation 2*

The purpose of the second test is to demonstrate the formation generation of the heterogeneous UAV–UGVs systems under the control of the NI formation tracking controller.

The communication topology between a master UAV (1) and two followers UGVs (2) is given in Fig. 23 and the dynamics of the three systems are expressed as in Tables 1 and 2; thus $\mathcal{Q}_a = [1\ 1;\ -1\ 0;0\ -1]$, $\mathcal{Q}_b = [0\ 0;\ -1\ 0;\ -1\ 0]$, and $\mathcal{Q}_r = [1;0;0]$. All four NI gain controllers are selected as (−2.4, −2.4, −2.4, −2.4, −5.5; −5.5). The geometry distances on the horizontal and vertical axes are given by letting $X_f = (X_{f_1}, X_{f_2})$. In there, $X_{f_1} = (100;50)\ (100;50)\ (100;50)\ (100;0)$ and $X_{f_2} = (-100;50)\ (-100;50)\ (-100;50)\ (-100;0)$. The master is driven along a rectangular trajectory which connects the four vertices: (80;100), (0;100), (0;0), (80;0).

As shown in Figs. 24 and 25, although the relative position references between the leader UAV and the slaves UGV is varied constantly, the desired formation based on consensus of the UAV–UGVs system is achieved. By imitating the master's behavior with the pre-defined distance offset $X_f$, the follower UGV 2 goes exactly through the four different nodes (−20;50), (−100;50), (−100; −50), (−20;0) while the follower UGV 3 travels around the points (180;50), (100;50), (180;0), (150;50). It is concluded that a time-varying formation for an homogeneous vehicles system, which contains different dynamics, can be obtained via our NI consensus protocol.

*9.1.3. Simulation 3*

Following the success of Simulation 1 and 2, the formation tracking and collision avoidance controller based on the NI theory is employed to a large-scale team of the UAV–UGVs robots exploring an unknown space in Simulation 3.

Most of the settings are kept as in Simulation 1. Fig. 26 indicates that the NI distributed cooperative tracking control law, along with the NI distributed obstacle avoidance method is able to drive gently and smoothly the six mobile robots from the initial positions (the lower left corner of (−150; −150) point) to the desired locations (the upper right corner of (400;400) point) in an inverted-V pattern along planned trajectories. Moreover, once a potential collision with obstacles is detected, the two proposed approach also guide the robots to steer around two stationary obstacles located at the facing sides before their predetermined formation shape is restored. The total time to complete traveling is only 8 s. Based on the achieved result, it is clear that the proposed approach also adapts to larger numbers of vehicles without increased complexity in communication requirements.

*9.1.4. Simulation 4*

Generally, APF is a powerful method of navigation for mobile robots in a cluttered environment. The key idea is to find a complete path from the starting location of the robot to the target location and simultaneously avoiding obstacles along its way. Despite the advantages that it offers, this method is not free from local minima and oscillation problems. The success of the control algorithm depends on the attractive and repulsive potential gradients acting on the robot. While the former part yields the attractive forces pulling the robot to goal, the latter part offers the repulsive forces pushing the robot away from obstacles.

In the final case, the performance of obstacle detection and avoidance using the NI control protocol compared to that of the APF as a benchmark. To do so, we recall the simulation settings

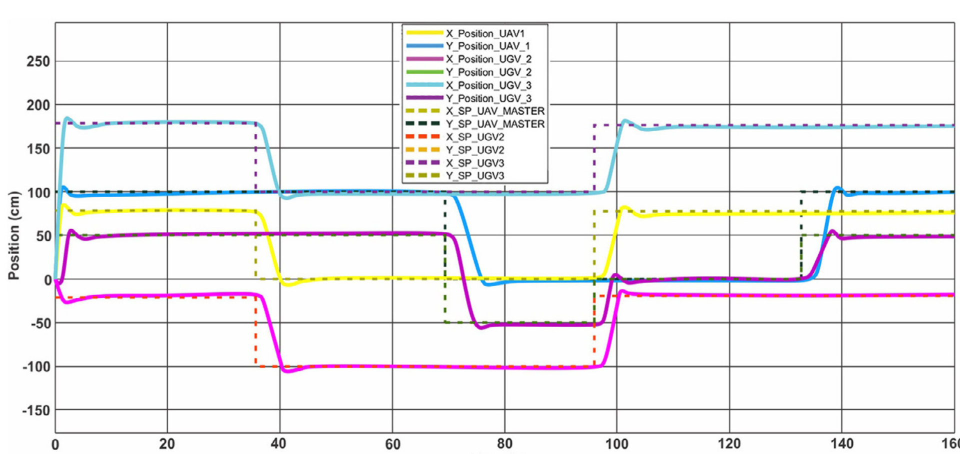



**Fig. 24.** Position response of the UAV–UGVs system over time.

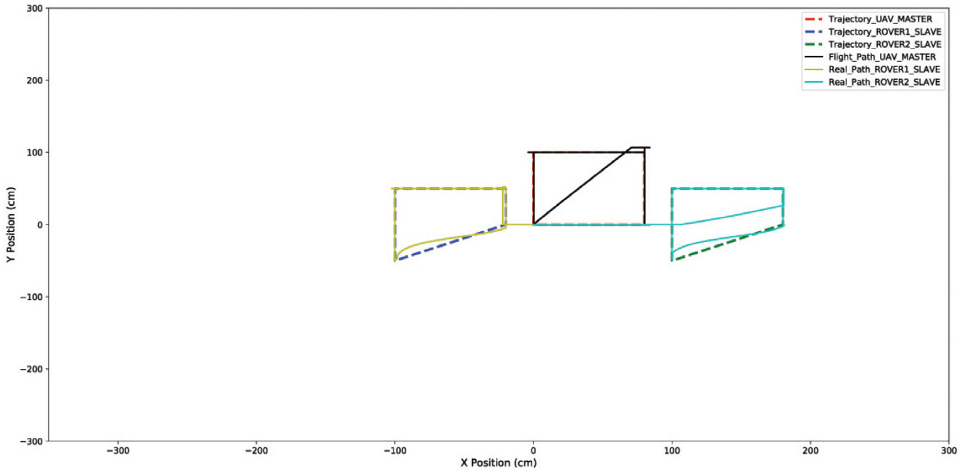



**Fig. 25.** The three polygons drawn by the UAV–UGVs position consensus.

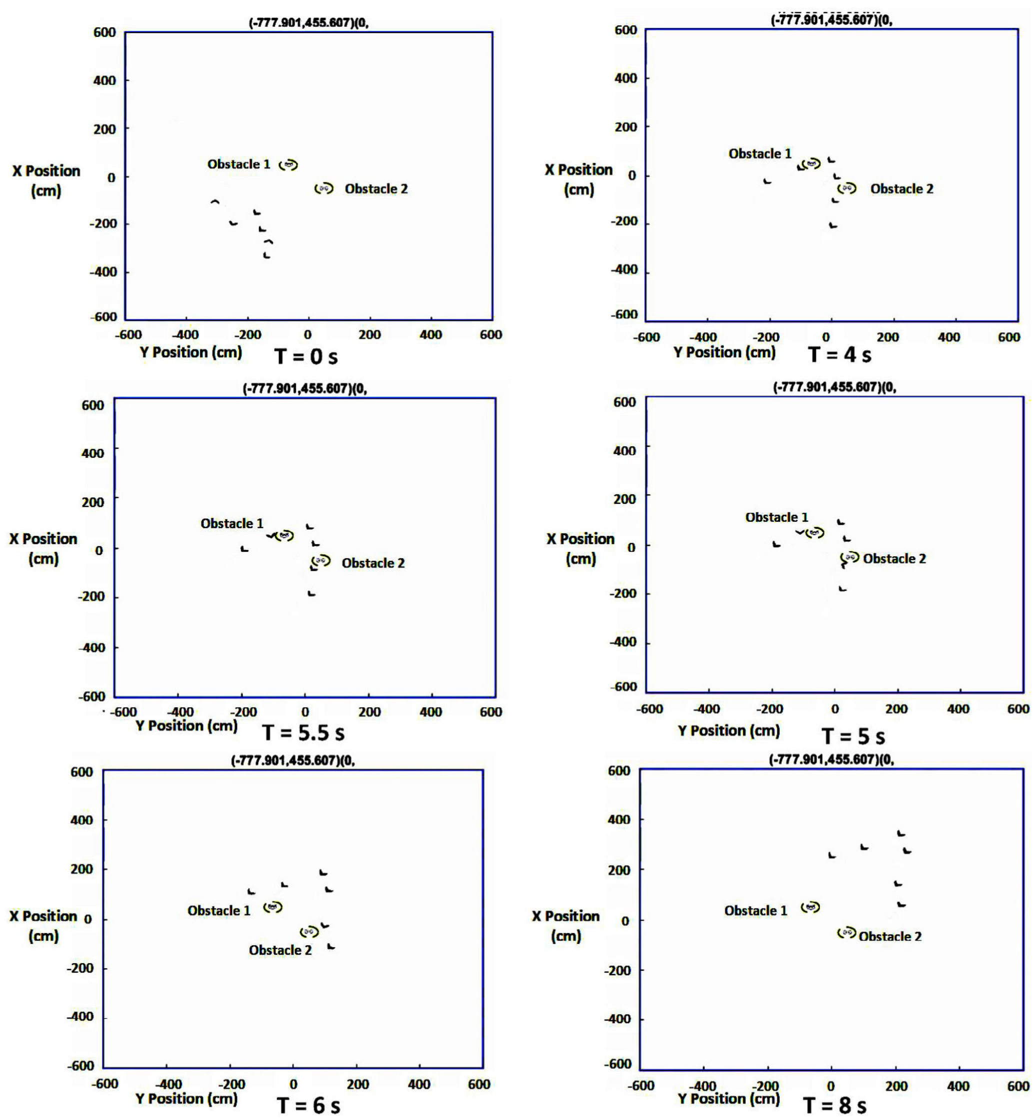
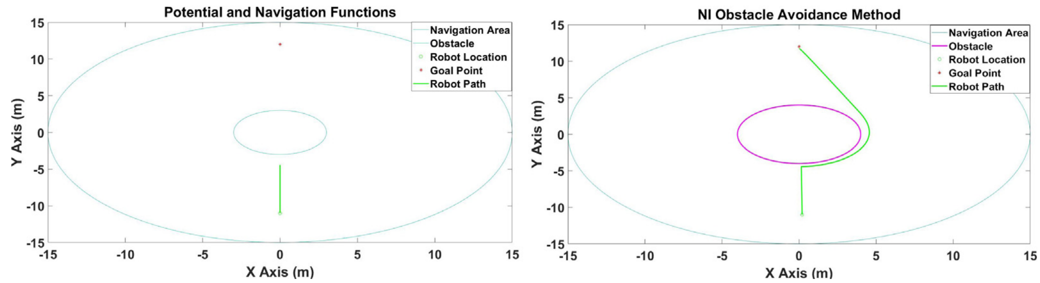
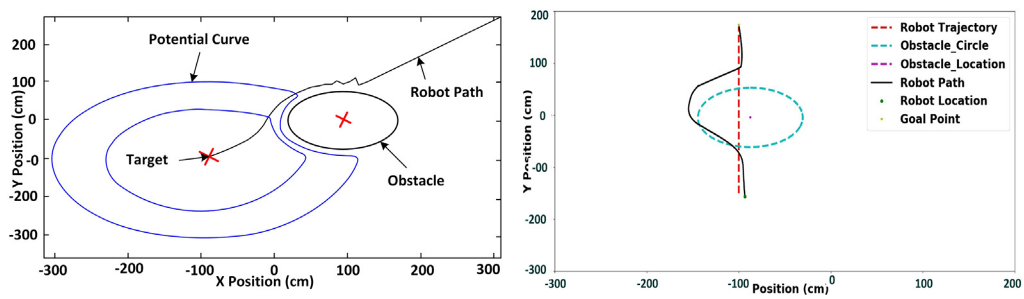



**Fig. 26.** Two facing obstacles avoiding process of the UAV–UGV systems using the NI approach.




(a) $APF$ (b) $NI$

**Fig. 27.** Local minimum case: (left) APF fails to find the path, (right) NI obstacle avoidance succeeds in finding the path.




(a) $APF$ (b) $NI$

**Fig. 28.** Disturbance case: (left) oscillatory trajectory using APF, (right) smoothness of the trajectory using NI strategy.

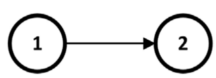



**Fig. 29.** Graph for communication of the two SNI systems.

in Choset, Hutchinson, Lynch, Kantor, Burgard, Kavraki, and Thrun (2005) as follows: the scaling factor $\zeta = 0.5$; the threshold value $dgoal = 15$; and a proximity threshold $Q = 1$. Two different tests are implemented to compare the performance of two algorithms. The radius of boundary sphere of an obstacle is 4.5 m.

In case of local minima, APF occasionally fails to find the path since net forces acting in opposite directions but existing on the same line cancel each other out, meaning that the robot gets stuck at that point. This problem cannot be predicted prior to prevent (Fig. 27(a)). Compared to APF, local minima problems are never experienced in our method since the repelling force is computed by the intersection area which has not local minima points (see Fig. 27(b)).

In case of oscillations in trajectory, Karnika and Indrani (2014) state that swings in the path of the robot associated with APF have not yet been completely solved, especially in narrow passages between a cluster of obstacles. In his work, oscillations were only reduced using the modified Newton's method in a cluttered workspace. As can be seen in Fig. 28(a), the contour of the potential surface is no longer circular due to the obstacle resulting in a zigzag fashion. In our approach, the virtual repelling force decreases gradually and linearly due to the reduction in the overlap amount between the heading vector and the obstacle circle when the robot moves far away from the obstacle. The trajectory is further smoothed out after applying our method (Fig. 28(b)).

Through the extensive comparisons between two control laws, it is demonstrated that the performance of NI controller exceeds the performance of APF function. Overall, the efficiency of NI method has been verified, especially in generating the smoother trajectories with no oscillations as well as avoiding local minima traps.

### 9.2. Experimental results

Based on the NI FA structure and the cooperative strategy discussed earlier in Sections 5 and 6, all three situations of collision avoidances using the UAV–UGVs/UGVs collaborative systems are tested in indoor environment containing some unexpected stationary obstacles.

In cooperative scheme of the two agents, the communication topology is given in Fig. 29 and thus $\mathcal{Q}_a = (1; -1)$, $\mathcal{Q}_b = (0; -1)$, $\mathcal{Q}_r = (1; 0)$. In cooperative scheme of the three agents, the communication topology is illustrated in Fig. 23; thus $Q_a = [1\ 1; -1\ 0; 0\ -1]$; $Q_r = (1;0;0)$. Both fundamental parameters are expressed as: $(X_r;Y_r) = (-100; -150)$ $(-100;170)$, $(X_f;Y_f) = (-200;0)$ for Case 1 and Case 3 and $(X_f;Y_f) = (-100;100)$ $(100;100)$ for Case 2, $V\_r_{max} = V\_ro_{max} = 30$ cm/s, $r = 57$ cm. The dimension of each packing box is 36cmL×36cmW×36cmH. The destination of lengthened trajectory on vertical axis *offdisy 3* is calculated as: *a[1] + r + 6* while that of the restoration trajectory *offdisy 4* is *a[1] + r + 38*. $k = (kx;ky) = (0.225;0.225)$ for the rovers and (0.35;0.35) for the UAVs. Similarly, *max_ahead* equals to 71 cm for the rovers and 100 cm for the UAVs. In addition, after tuning of gains for the tracking and consensus performance, both NI controllers are chosen as $(-0.0028; -0.0028)$ for the rovers and $(-0.7; -0.7)$ for the UAVs.

During the experiments, either robot occasionally collides with the in-front obstacle while it is moving towards the obstacle circle's center. This is due to inaccurate measurement of the velocity sensor, resulting in a slight fluctuation in the $\overrightarrow{ahead}$ vector's direction around the center point. Because of this constant change in every sample time, the average value of all steering forces is nearly 0. Therefore, no effects will be produced on the corresponding robot. As shown in Fig. 30, by making the $\overrightarrow{ahead}$ vector's direction depend on the $\vec{des}$ vector, this problem can be

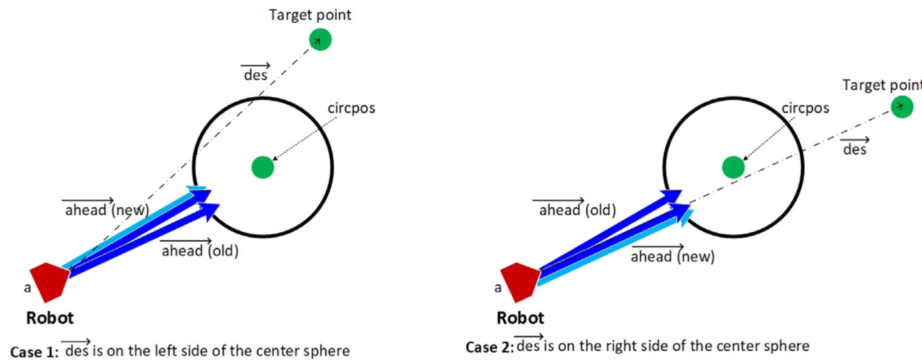



**Fig. 30.** Direction altering of the ahead vector.

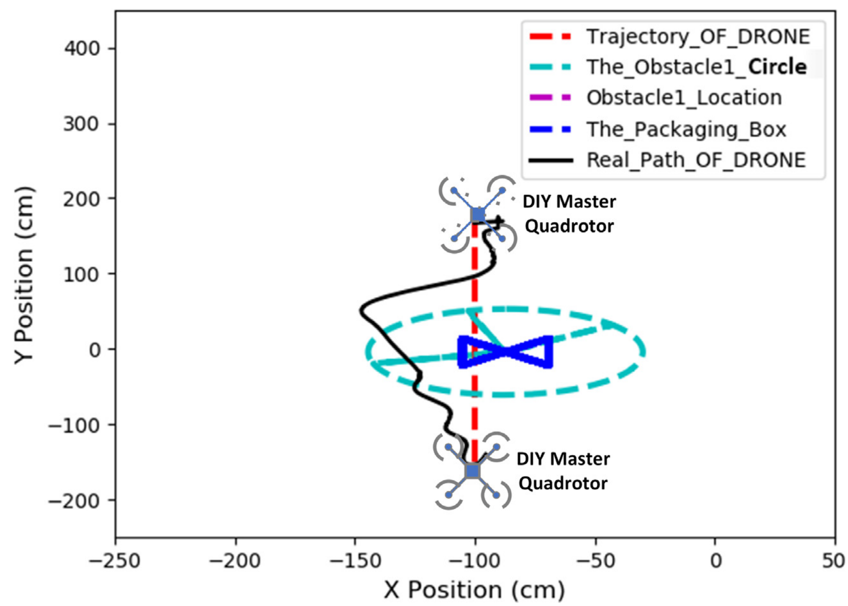



**Fig. 31.** Obstacle collision avoidance via the NI systems theory.

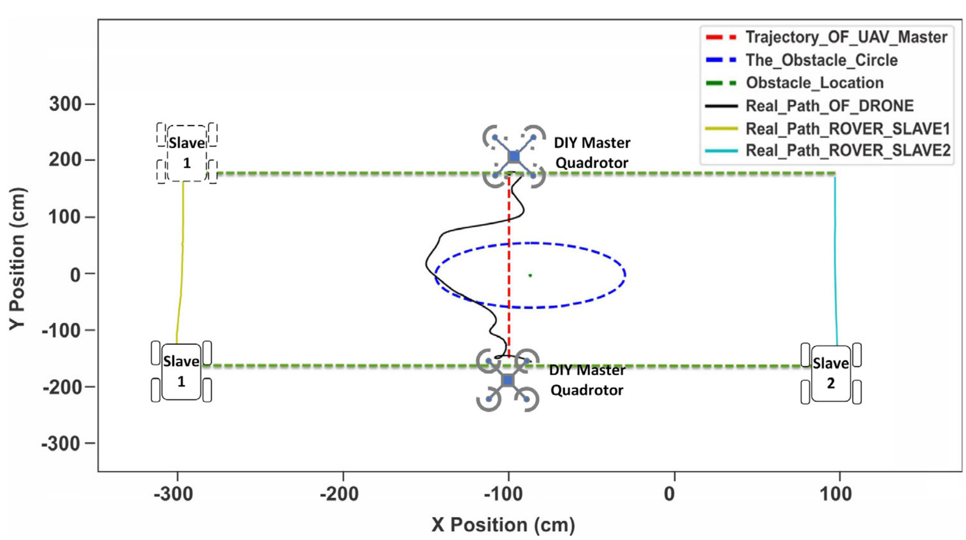



**Fig. 32.** Consensus-based NI cooperative formation control with the NI master–obstacle collision avoidance for a UAV–UGV system in Case 1.

tackled effectively. As illustrated in Fig. 35, the follower is still able to evade the obstacle and move towards target although the robot heading is in a straight line with the obstacle center.

Fig. 31 depicts the outcome of a real flight test with obstacle avoidance. Based on the distributed NI obstacle avoidance technique, once a possible collision with the static packing box is

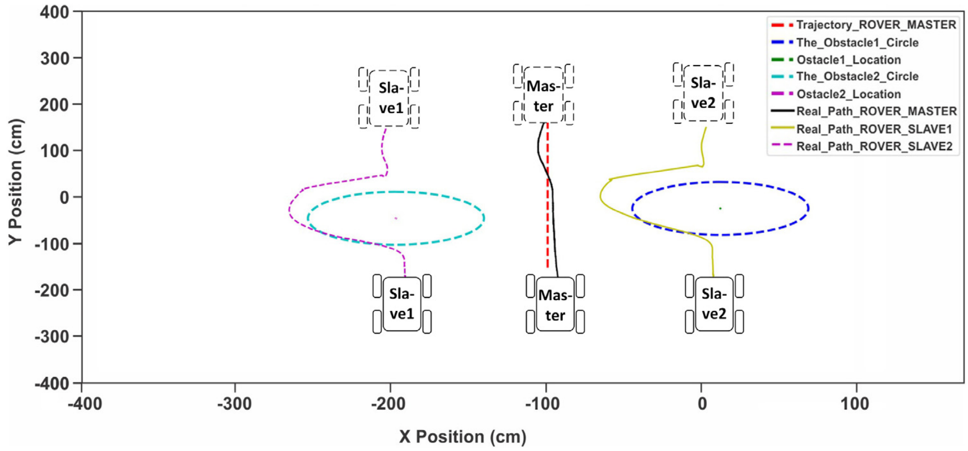



**Fig. 33.** Consensus-based NI horizontal-line formation control with the follower–obstacle collision avoidance for a UGV–UGV system in Case 2.

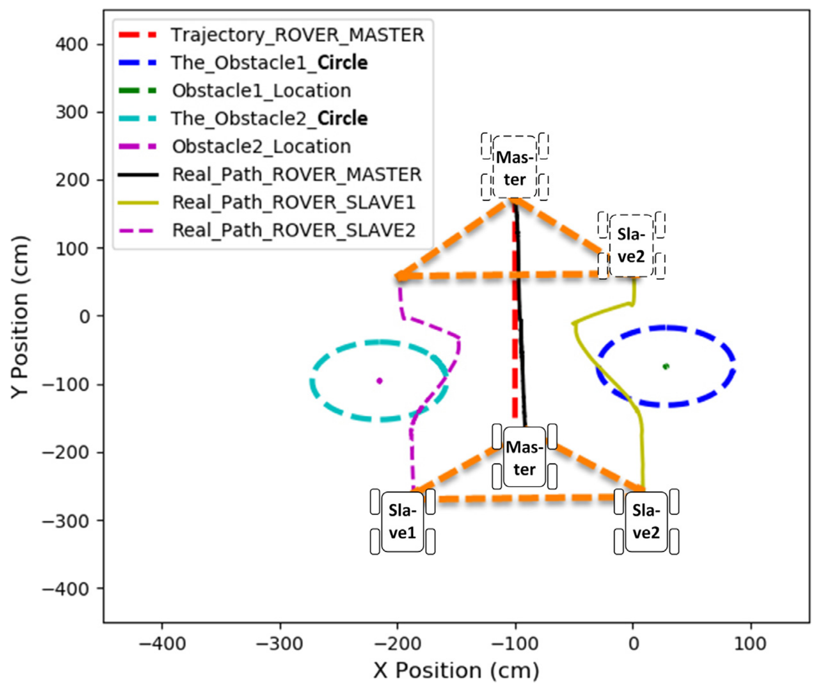



**Fig. 34.** Consensus-based NI triangle-shaped formation control with the follower–obstacle collision avoidance for a UGV–UGV system in Case 2.

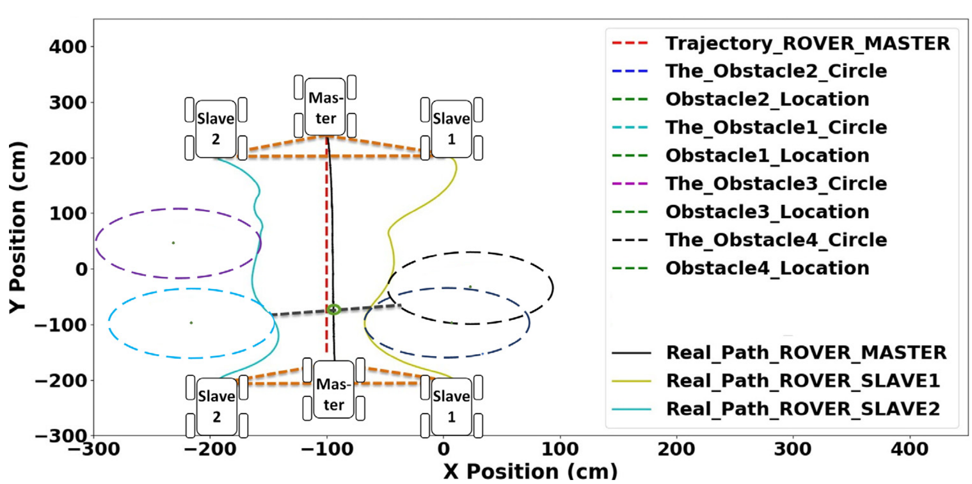



**Fig. 35.** Obstacle avoidance for the multi-UGV system in an uncertain environment.

predicted, the UAV is immediately guided to follow a real-time safe path (the obstacle avoiding trajectory) relating to its escape velocity. After the obstacle is crossed through, the UAV restores its path to the initial one and then quickly flies to the target point (see Fig. 34).

After the performance of the single UAV in the first obstacle avoidance experiment was satisfactory, a series of three successive experiments using a multi-robot collision avoidance system, which are classified through the three cases of obstacle avoidance, were conducted. Obtained results in Case 1 are depicted in Fig. 32, while those in Case 2 is indicated in Fig. 33. Finally, a whole situation, in which four obstacles are placed randomly in a completely-unknown environment, as shown in Fig. 35, was set up to validate the efficiency of our proposed methods.

It is noted that the position output feedback consensus for a collaborative team, including the two different heterogeneous systems (UAV–UGVs and UGV–UGVs) is guaranteed via the NI control algorithm in Tran et al. (2017). The robots are in perfect formation, including horizontal line and triangular pattern. As can be seen in all plots, the maximum position consensus error is approximately 7 cm. Moreover, by introducing another NI distributed control law for obstacle avoidance and a new cooperative strategy, any robots (the UAV or the UGV) at any positions in the connected formation (the leader or the follower) are capable of detecting and avoiding obstacles by steering correctly around the obstacle's virtual circle. Specifically, the maximum distance between the obstacle avoiding trajectory and the boundary of the virtual circle is roughly 4 cm in all cases.

On the other hand, formation regrouping after passing the obstacle is implemented quickly and effectively by the safe trajectory planner. Minimal oscillations are seen on the actual paths. Finally, since our mobile robots can pass through all obstacles and reach the target point in a given triangular formation as can be seen in Fig. 35, our obstacle avoidance and formation generation strategy based on the NI-systems control theory is very suitable for exploration applications where the environmental information is limited or unknown.

The experimental results reported in this paper are better than those presented by the authors in Antonelli et al. (2008), Dai et al. (2014), Nishio, Nonaka, and Sekiguchi (2017) and Lee and Chwa (2018). With respect to the APF utilized for path planning and inter-UAV collision avoidance in Abeywickrama et al. (2017) and Uyanik (2011), the obstacle avoiding trajectory of the robot planned by NI method is smoother and gentler as can be seen in all tests while maintaining minimal steady-state error with the desired reference. Furthermore, as mentioned in Section 5 of Antonelli et al. (2008) and Dai et al. (2015, 2014), the oscillation phenomena were seen on the UGV's traveling paths while suddenly switching from the move-to-goal task to the obstacle-avoidance one and vice versa. As presented in all our real tests, the robots' output trajectory is quite smooth throughout the avoiding process. This outstanding performance comes from our proposed cooperative approach that determines the various missions for each robot and the safe trajectory generation that combines the obstacle avoidance task with the target-reaching task smoothly. Moreover, as used in Dai et al. (2015, 2014) and Lee and Chwa (2018), the traditional GOACM method computes the escape angle for the UGV based on the complex shape of the obstacle, which would not work with complex-shaped obstacles. To deal with this issue, the obstacle's virtual circle was employed in our proposed method. Moreover, our solution is also unobstructed by the dead-lock angle problem as occurred in Nishio et al. (2017) since the escape velocities on the $x$ and $y$ axes are used to avoid obstacles. Another approach named enhanced APF, which was presented in Nishio et al. (2017), shown its ability to avoid mobile obstacles based on prediction of obstacles' future motion and the fuzzy output-feedback control. As displayed in our simulation results, our enhanced method not only shows the same capability but also takes into account the control stability for the whole architecture, involving tracking performance, formation preservation and static/mobile obstacle avoidance. On the other hand, our FA solution can also be applied to even heterogeneous systems; e.g. the system of UAV and UGVs combination, which contains different dynamics whilst most of the recent studies concentrated on the consensus and obstacle avoidance for only homogeneous systems such as UGVs' system (e.g. Dai et al., 2014; Lee & Chwa, 2018; Nishio et al., 2017) or UAVs' system (e.g. Dong et al., 2017; Zhang, 2017). Some cooperative control scenarios for the UAV–UGVs system were presented in Section 8 to prove the distinctive feature of our theory. Finally, it is worth nothing that under the control the NI control strategies, the high order and complex systems such as the UAV or the UGV still accomplish desired goals as successfully as the lower system does. This positive result offers an enormous opportunity for the high-order multi-agent formation control, which is a significant shortcoming of the previously proposed methods in Dong et al. (2017), Wang et al. (2018) and Wang et al. (2015).

Also note that only three novel collision avoidance scenarios for inhomogeneous mobile robots are taken into account in generating effective collision-free trajectories in an unstructured environment. This result is attained thanks to the obstacle self-avoidance technique that has been applied to each robot.

Last but not least, it is clear that our distributed NI formation control approach for Multi-Input and Multi-Output (MIMO) system has been effectively developed from the original consensus theory proposed by Wang et al. (2015). These improvements include a distance consensus architecture between the leader and followers, an NI obstacle avoidance approach, and a formation planner. Based on our novel technique, the robots are able to make a distance consensus follow a pre-defined formation and evade any obstacles at the same time.

The experimental tests performed can be viewed from the following video links: https://tinyurl.com/y8huzppg.

## 10. Conclusions

Robust cooperative control for both UAV–UGVs and UGV–UGV multi-robot systems is guaranteed via the distributed FA architecture and its implementation strategy. The vital contributions in our research are concluded as follows: (1) providing the homogeneous control structure for the formation and obstacle avoidance problem using the NI systems theory; (2) showing how consensus and obstacle avoidance problems can be implemented simultaneously via our novel cooperative strategy; (3) the distributed NI obstacle avoidance approach can be used to overcome all static and dynamic obstacles which are moving at various velocities; (4) proving the guaranteed stability of the whole FA architecture under unexpected effect of position noise using the NI systems theory; (5) presenting an active obstacle avoidance approach for multi-robotic system to work in unknown environments; (6) the NI systems theory widens the scope of practical applications and is adopted to many various types of cooperative system, e.g. the system of UAV–UGV and of UGV–UGV. Experiments for the NI formation control with obstacle avoidance have confirmed that the robots are able to achieve and maintain a specific desired shape of a multi-agent rigid formation along a time-parameterized path, while still to guarantee collision avoidance with both static and mobile obstacles. All performances meet fully the major goals.

Evidently, the performance of the NI obstacle avoidance exceeds the performance of the conventional potential field function control strategy mentioned in our simulation. Overall, the robot path produced by the NI obstacle avoidance is smooth and stable even while avoiding mobile obstacles and keeping minimal steady-state error.

Although only three robots have been used in simulation and indoor experiments to evaluate the feasibility of the NI theories, more significant numbers of autonomous robots can also be employed in real applications without concern about data dropping or time delay. It is because the agents know only their own states and the desired robot–robot and obstacle–robot relative positions (measured directly by each slave's machine vision) without having to share the state information of all robots at every time step as in Dong et al. (2016, 2017) and Huang, Teo, Liu, and Dymkou (2017). Also, to improve the efficiency of large-scale computations, F-A architecture is also carried out in a distributed way. As a consequence, our robots are still able to produce their normal behaviors when a big group is utilized. These distinctive characteristics of our two NI theories become practical as the vast number of agents are being coordinated in a hazardous environment where data communication between agents is limited.

Furthermore, our collision detection and obstacle avoidance approaches are only adopted in a 2D context. However, they may be extended for operating in a spatial environment by considering an intersection between the $\overrightarrow{ahead}$ vector and an obstacle's virtual sphere.

A shortcoming of our current methodologies is that the mutual collision avoidance between robots has not been addressed when the spacing between two obstacles is less than that of two robots. In this case, the only way for the robots to pass through the obstacles is to change their formation shape. In future research, we will execute a queuing behavior in which all robots are rearranged into a line-column formation before passing through tightly spaced obstacles.

## CRediT authorship contribution statement

**Vu Phi Tran:** Conceptualization, Investigation, Methodology, Testing, Software, Writing. **Matthew A. Garratt:** Conceptualization, Supervision, Validation, Reviewing and editing. **Ian R. Petersen:** Conceptualization, Supervision, Validation, Reviewing and editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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